POLYMERIC MATERIALS

Polymers – basic concepts

Terminology:

- **Polymer** – "Poly" meaning "many" and "mer" meaning "unit".
- **Monomers** – Basic building blocks of a polymer.
- **Macromolecule** – extremely large collections of molecules to form one polymeric chain .
- **Plastics** – a synonym for polymers.

- The word "plastic" comes from the Greek word "plastikos", meaning capable of being molded and shaped.
- Polymers can be natural or synthetic (man-made)
- The earliest polymers, such as cellulose, were made from natural organic materials from animals and vegetable products

Most commonly, polymers are constituted by a **hydrocarbon backbone (chains formed by C and H)** to which different functional groups may be inserted. However, other structures exist, too; for example, silicon (Si) can form **polysiloxanes** (**silicones**), based on –Si-O-Si-O-Si- chains; they are widely used as sealants.

Table 14.1 Compositions and Molecular Structures for Some of the Paraffin Compounds: C_nH_{2n+2}

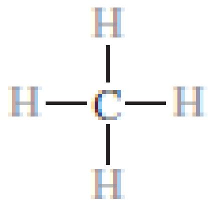
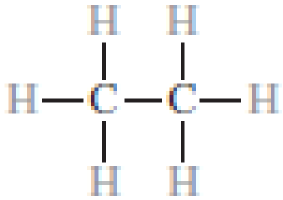
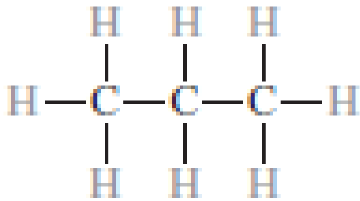
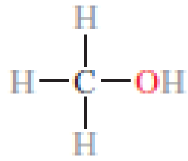
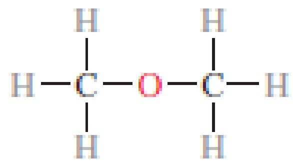
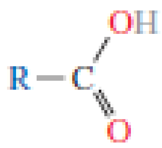
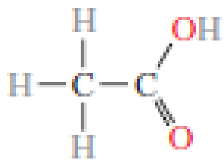
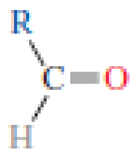
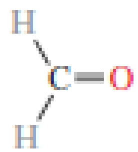
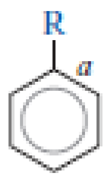
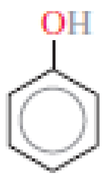
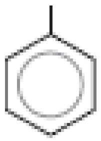
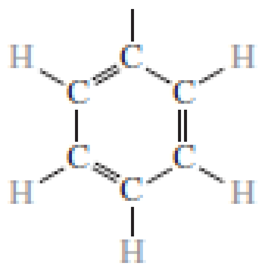
Name	*Composition*	*Structure*	*Boiling Point (°C)*
Methane	CH_4		−164
Ethane	C_2H_6		−88.6
Propane	C_3H_8		−42.1
Butane	C_4H_{10}		−0.5
Pentane	C_5H_{12}		36.1
Hexane	C_6H_{14}		69.0

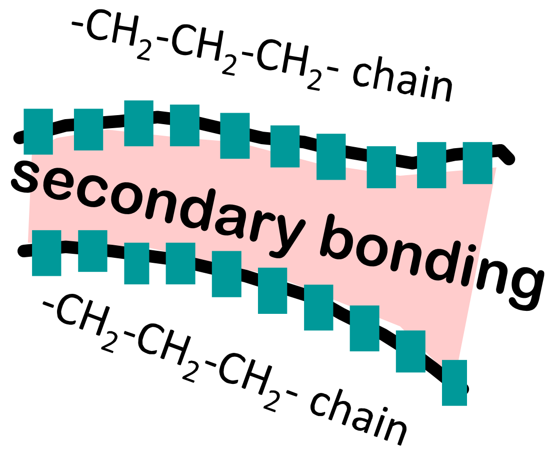
Table 14.2 Some Common Hydrocarbon Groups

Family	*Characteristic Unit*		*Representative Compound*
Alcohols	R—OH		Methyl alcohol
Ethers	R—O—R′		Dimethyl ether
Acids			Acetic acid
Aldehydes			Formaldehyde
Aromatic hydrocarbons			Phenol

[a] The simplified structure  denotes a phenyl group, 

Advantages of using polymers:

- low cost
- low electrical and thermal conductivity (insulation)
- low density
- high strength-to-weight ratio
- resistance to chemical corrosion
- noise reduction
- assortment of colors (aesthetics)
- ease of manufacturing
- minimal additional surface treatments
- forms of availability such as: tubes, films, sheets, plates, rods, etc.



Additives can be used:
fillers, lubricants, pigments…
Ex.: carbon black (in tires)

Limitations

- Low mechanical properties
- Low thermal resistance
- Flammability
- Damage due to visible light and UV

Polymerization

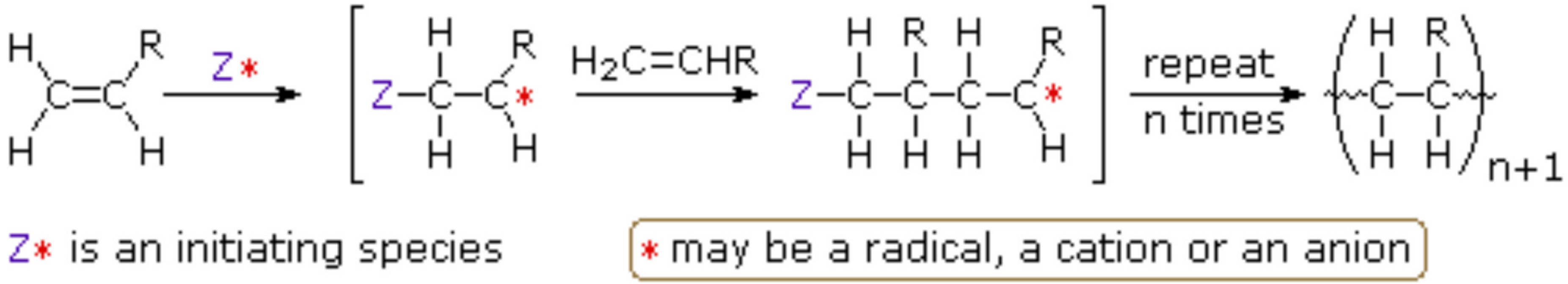
Polymerization is the process of combining many **monomers** into a covalently bonded chain or network.

ADDITION POLYMERS

All the monomers from which addition polymers are made are alkenes or functionally substituted alkenes (alkenes have at least one C = C bond). Example: polyethylene, derived from the ethylene ($H_2C = CH_2$).

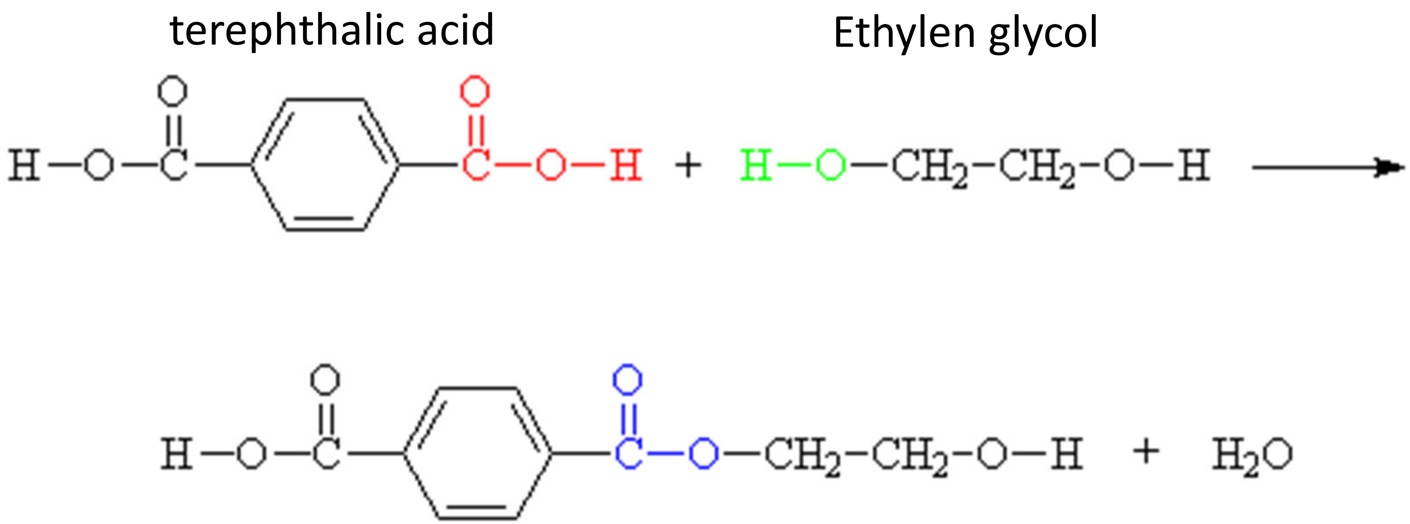
Many of poly-addition reactions are known to proceed in a stepwise fashion by way of reactive intermediates. A general diagram illustrating this assembly of linear macromolecules (**chain growth polymers**) is shown below.

Note that an initiating species, i.e. a catalyst (indicated as Z*), is necessary so that the reaction can run.



CONDENSED POLYMERS

In polycondensation, polymerization proceeds by **functional group transformations**; it typically involves the **loss of a small by-product**, such as water, and **combines two different monomers** in an alternating structure. Example: polyethylenterephtalate (PET):



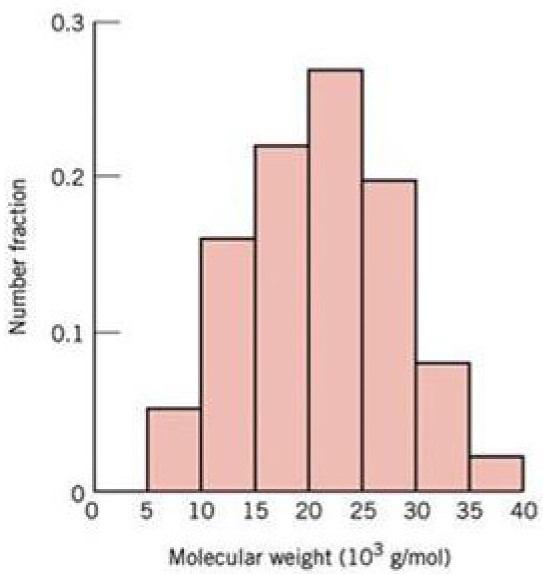
Condensation polymers form more slowly than addition polymers, often requiring heat. The terminal functional groups on a chain remain active, so that groups of shorter chains combine into longer chains in the late stages of polymerization. The presence of polar functional groups on the chains often enhances chain-chain attractions, particularly if these involve hydrogen bonding, and thereby crystallinity and tensile strength are increased.

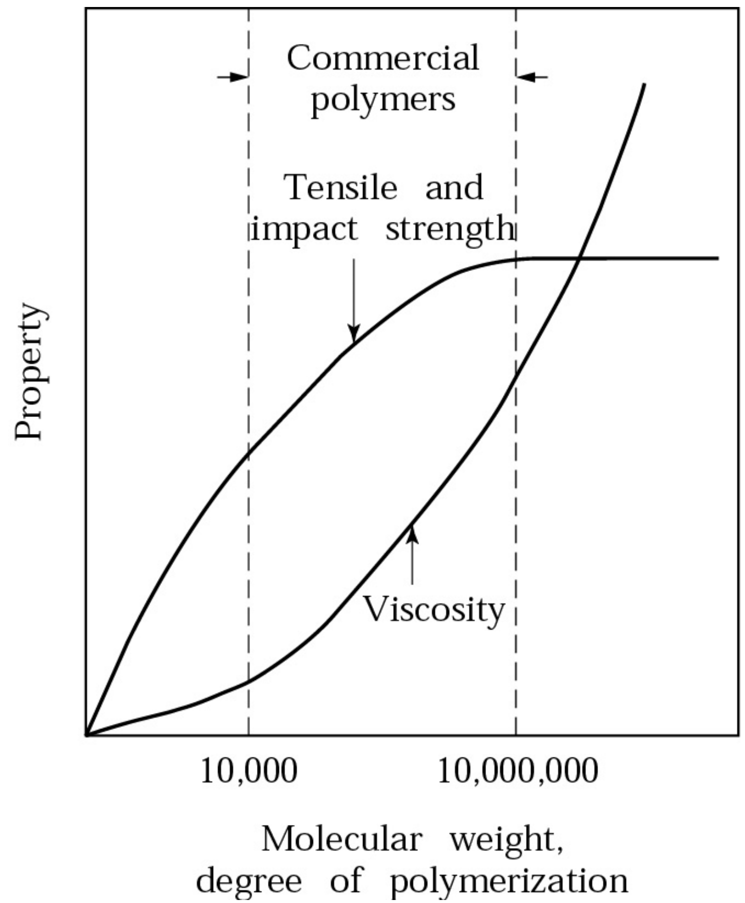
Molecular weight

• Molecular Weight Distribution (MWD), is the sum of the molecular weights of the mers in a macromolecule (polymer chain)

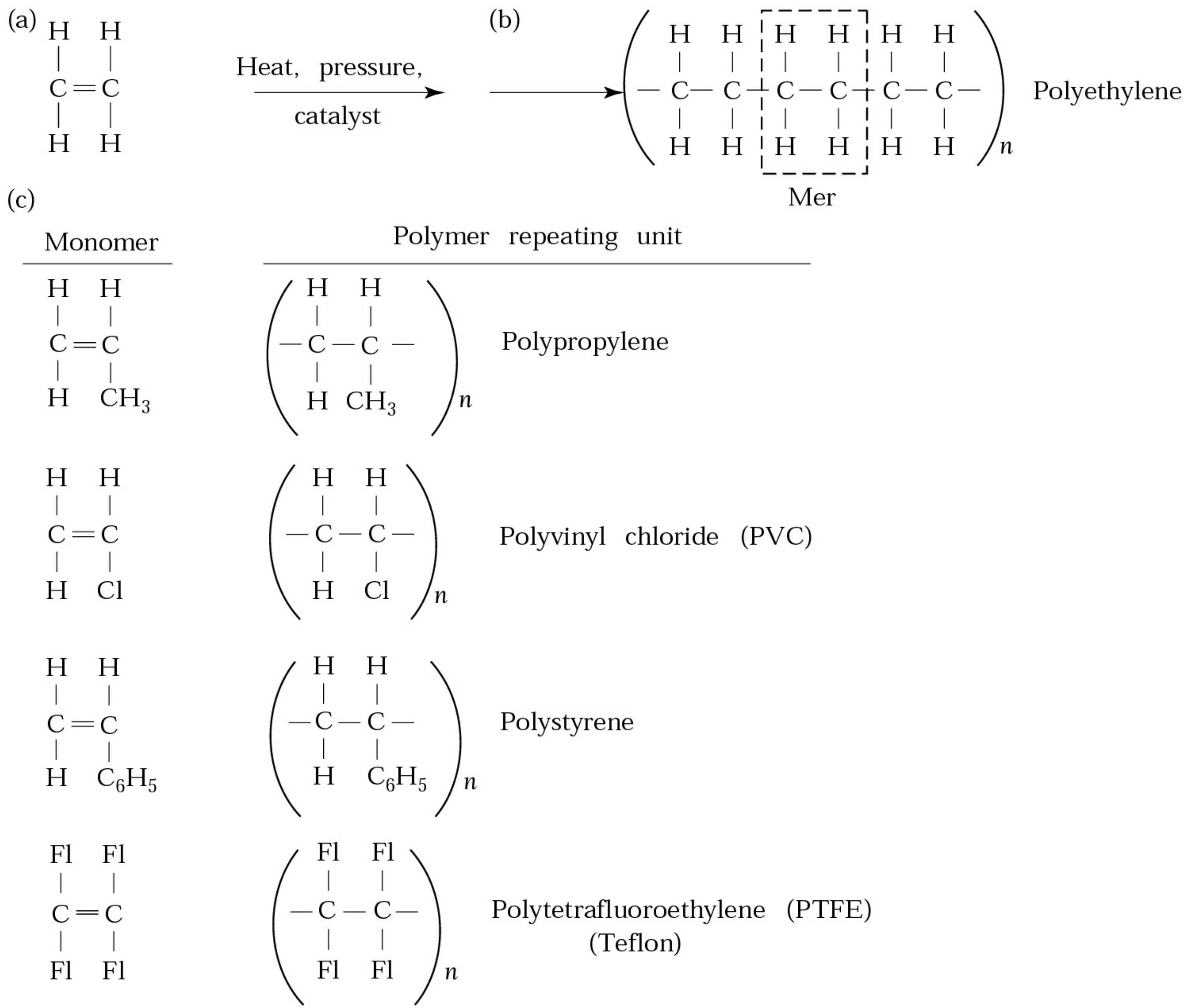
• Degree of Polymerization (DP) is the size (length) of the polymer chain

MWD and DP strongly affect the mechanical properties and viscosity of polymers





Composition



Hydrocarbon Functional Groups

Class	Functional Group		General Formula
Alcohol	hydroxyl group	-O — H	R - OH
Alkyl halide or halocarbon	halogen	— X	R — X
Ether		— O —	R — O — R′
Aldehyde	carbonyl group with H	O || — C — H	O || R — C — H
Ketone	carbonyl group	O || — C —	O || R — C — R′
Carboxylic acid	carboxyl group	O || — C — OH	O || R — C — OH
Ester		O || — C — O—	O || R — C — O — R′
Amine (also see amide in table 26.1)	amine group	| —N—	R′ | R — N — R″

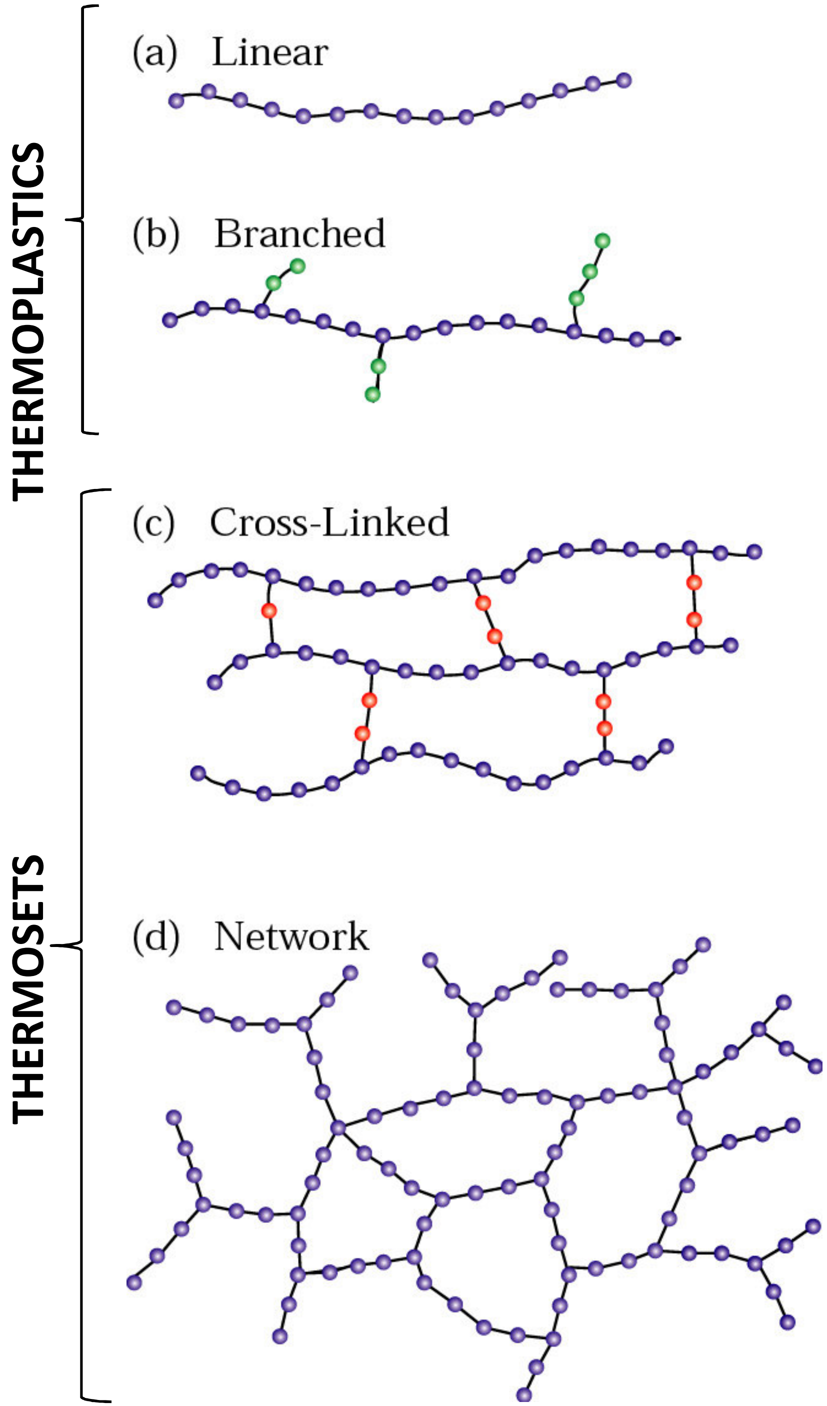
Structure

- Linear Polymers
  - ➢ Sequential structures

- Branched Polymers
  - ➢ Increased resistance to deformation and applied loads.

- Cross-Linked Polymers
  - ➢(Thermosets) higher hardness, strength, stiffness, brittleness, and better dimensional stability.

- Networked Polymers
  - ➢(highly cross-linked), have a higher resistance when exposed to high energy radiation, UV light, X-rays, or electron beams, as well as higher mechanical properties .

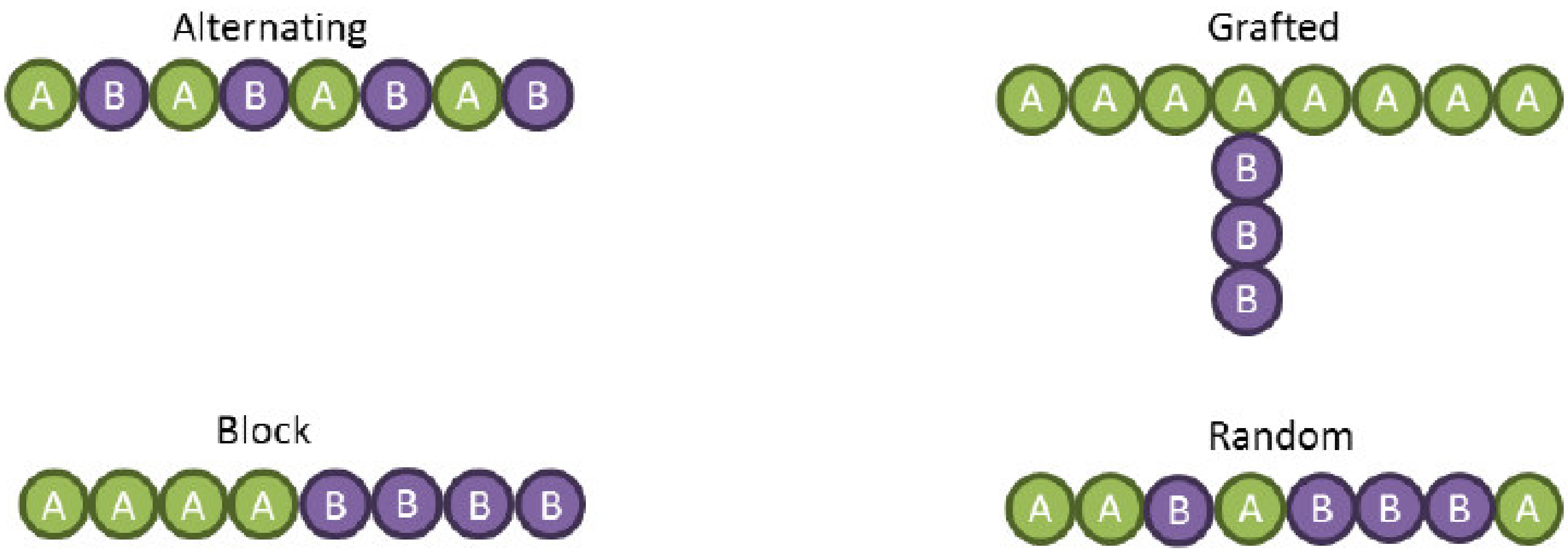


- **Copolymers** contain two types of polymers
  - ➢ Ex: Styrene-butadiene, used in making tires

- **Terpolymers** contains three types of polymers
  - ➢ Ex: Acrylonitrile-butadiene-styrene, used to make helmets

Purposes:

- Take the «best» from the constituing polyemers
- Modulation of properties



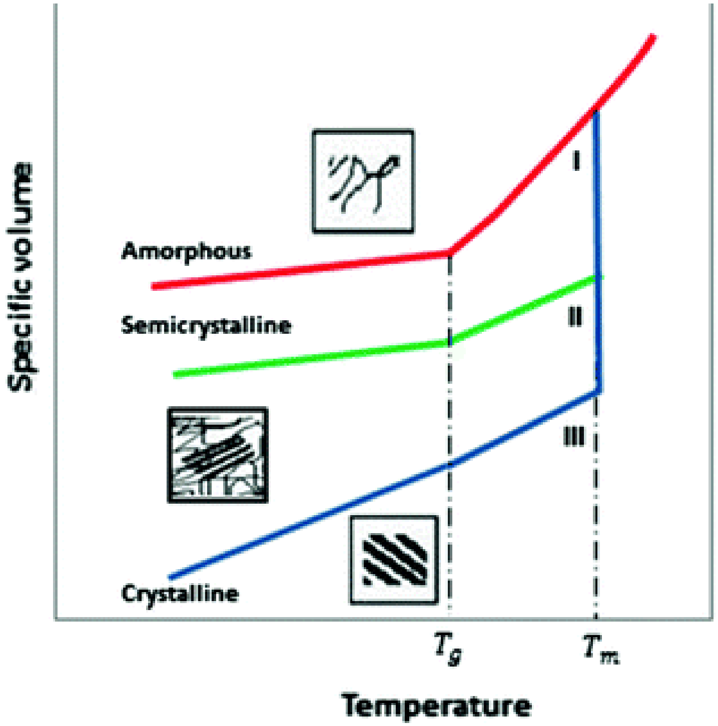
Crystallinity

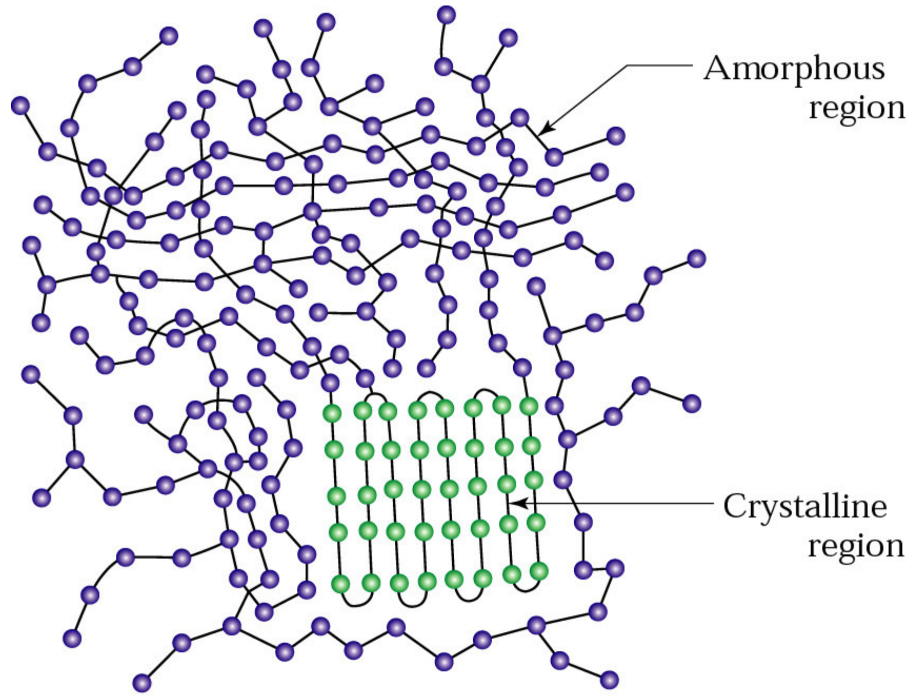
Polymers can be:

• **Amorphous**, the polymer chains exist without order (e.g. PVC).

• **Semi-crystalline** → crystallites: the regions arrange themselves in an orderly manner. PTFE can reach 98% crystallinity

Crystallinity can be increased by controlling (decreasing) the rate of solidification during the cooling process





Folded-chain model

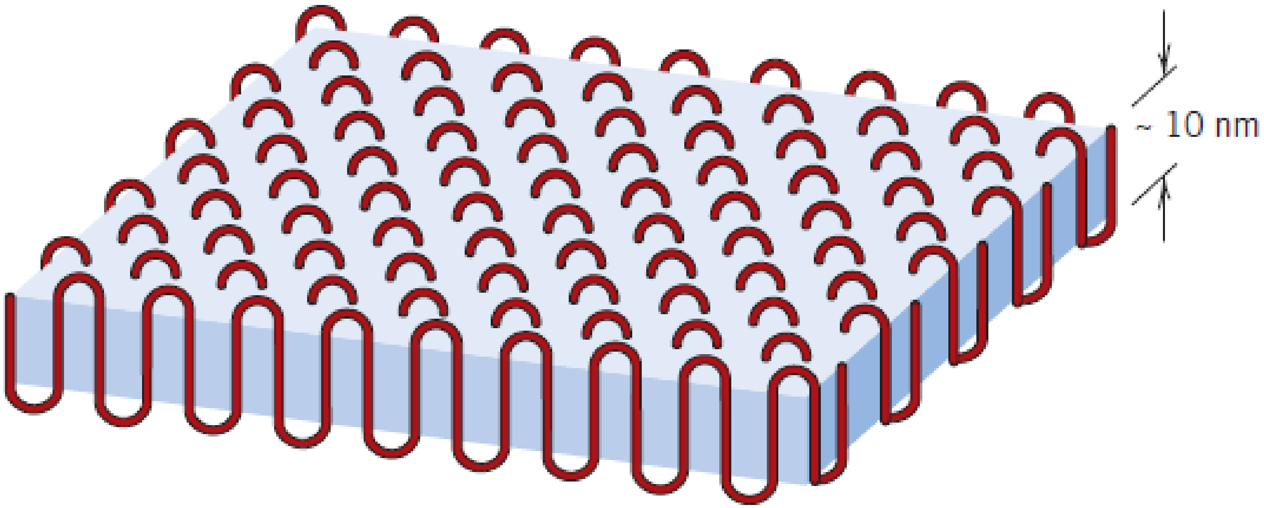
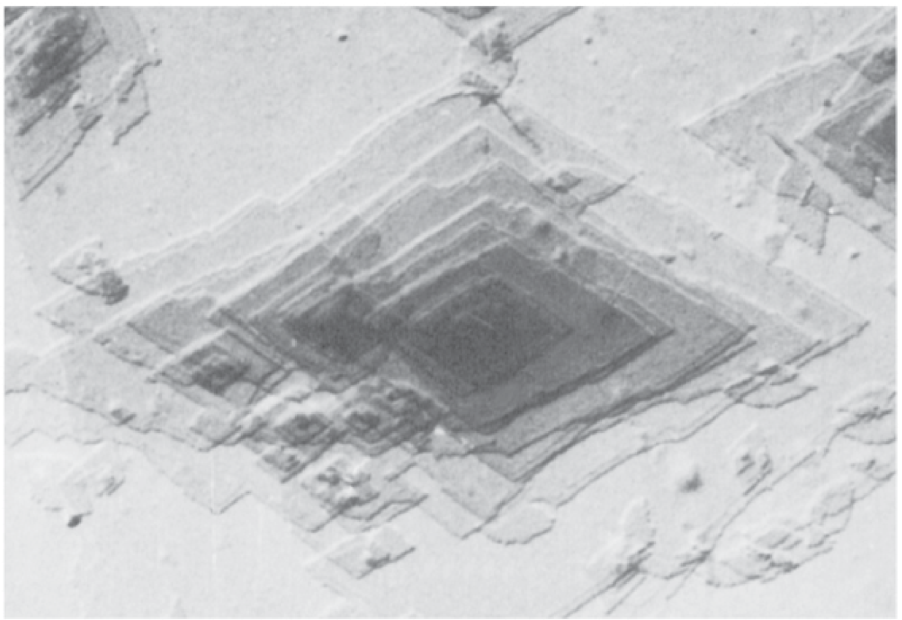


Figure 14.12 The chain-folded structure for a plate-shaped polymer crystallite.

Figura 14.11 Micrografia elettronica di un singolo cristallo di polietilene. 20000×. [Da A. Keller, R.H. Doremus, B.W. Roberts and D. Turnbull, (Editors), *Growth and Perfection of Crystals*, General Electric Company and John Wiley & Sons, Inc., 1958, p. 498].



Polymer crystalline structure in 3D: ***spherulites***

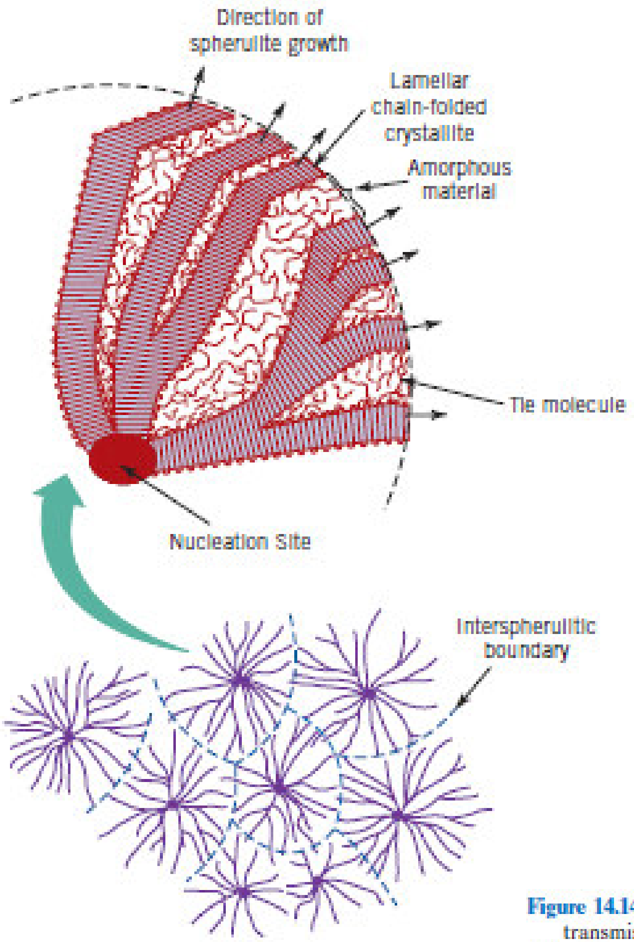
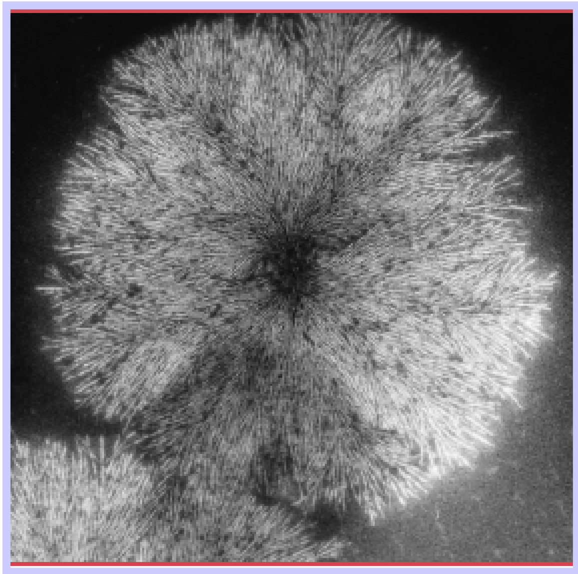


Figure 14.13 Schematic representation of the detailed structure of a spherulite.



Spherulite, natural rubber

Figure 14.14 A transmission photomicrograph (using cross-polarized light) showing the spherulite structure of polyethylene. Linear boundaries form between adjacent spherulites, and within each spherulite appears a Maltese cross. 525×. (Courtesy F. P. Price, General Electric Company.)

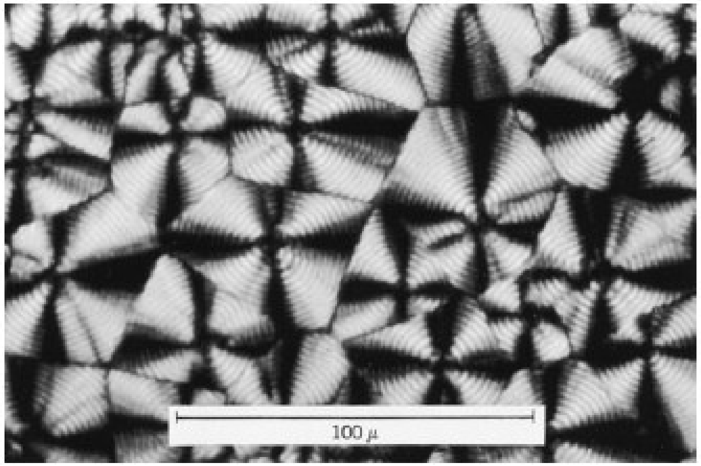
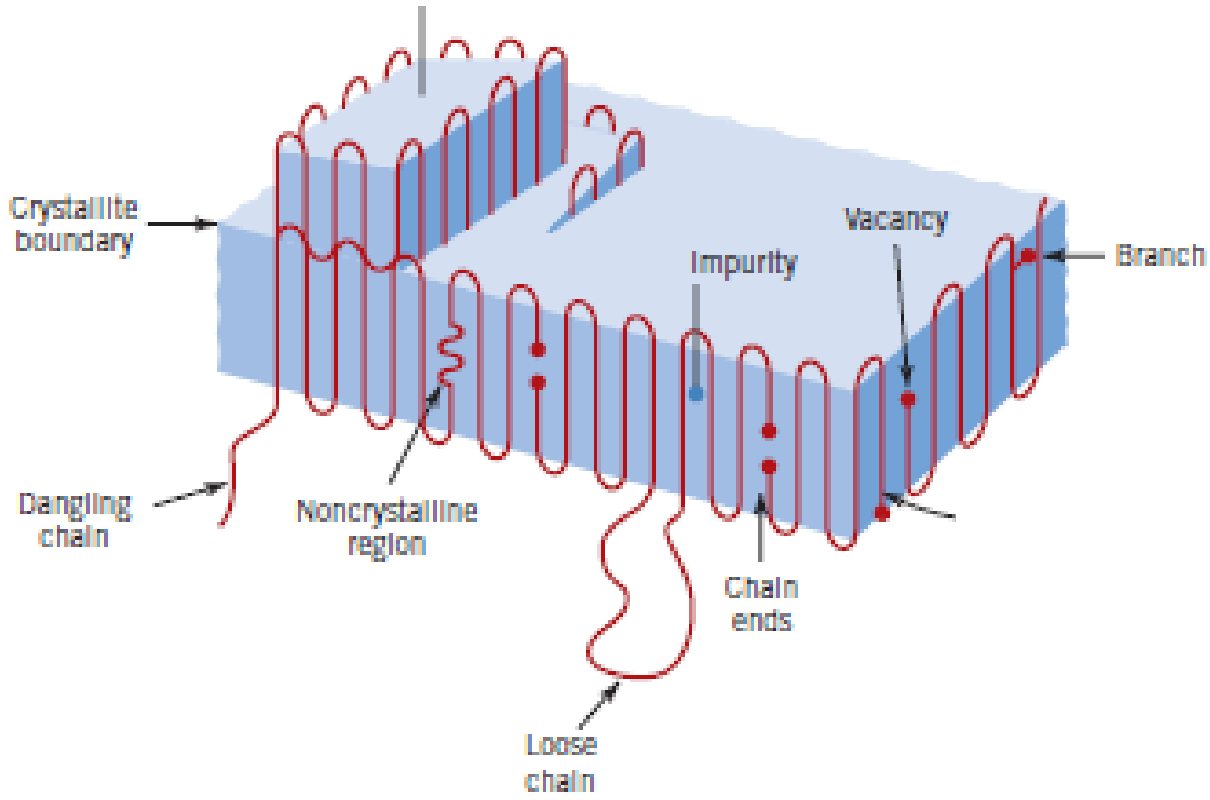


Figure 14.15 Schematic representation of defects in polymer crystallites.



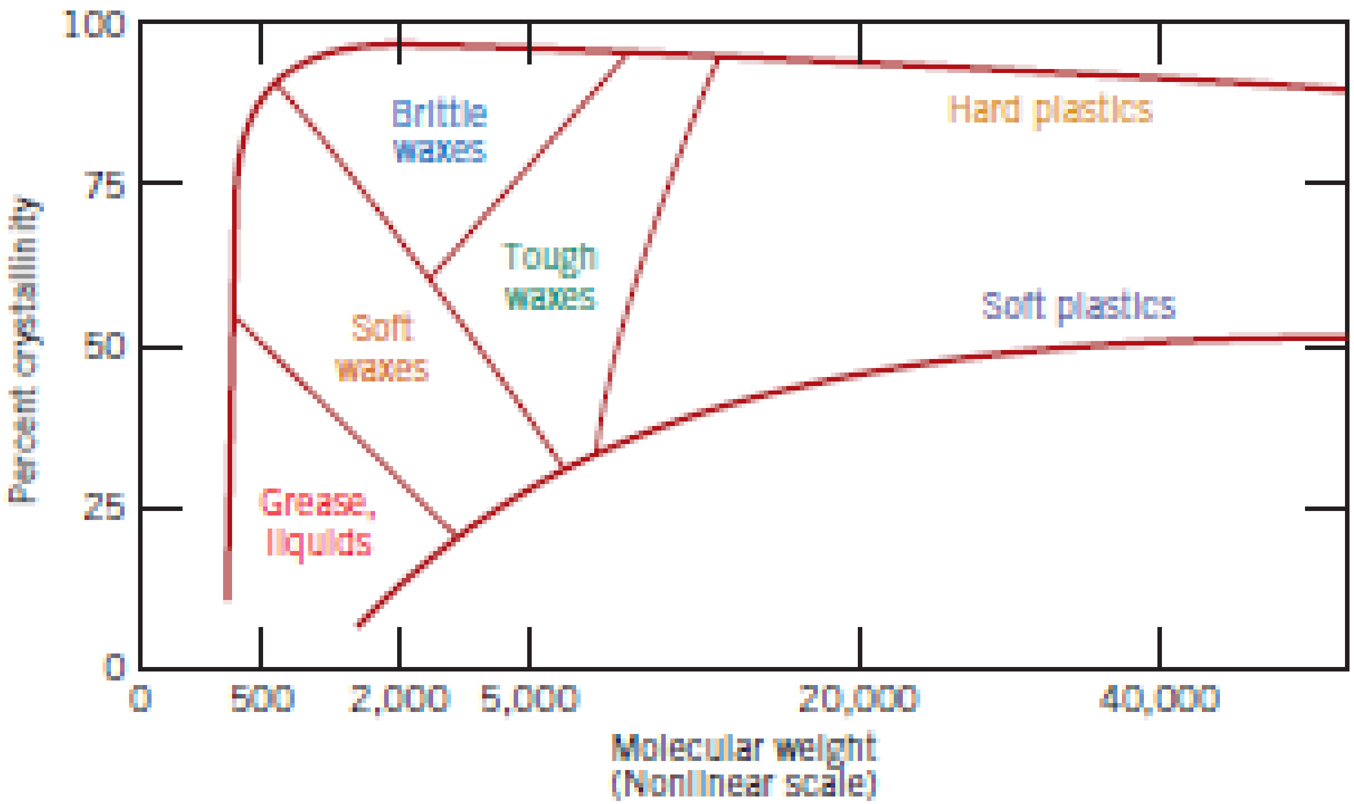
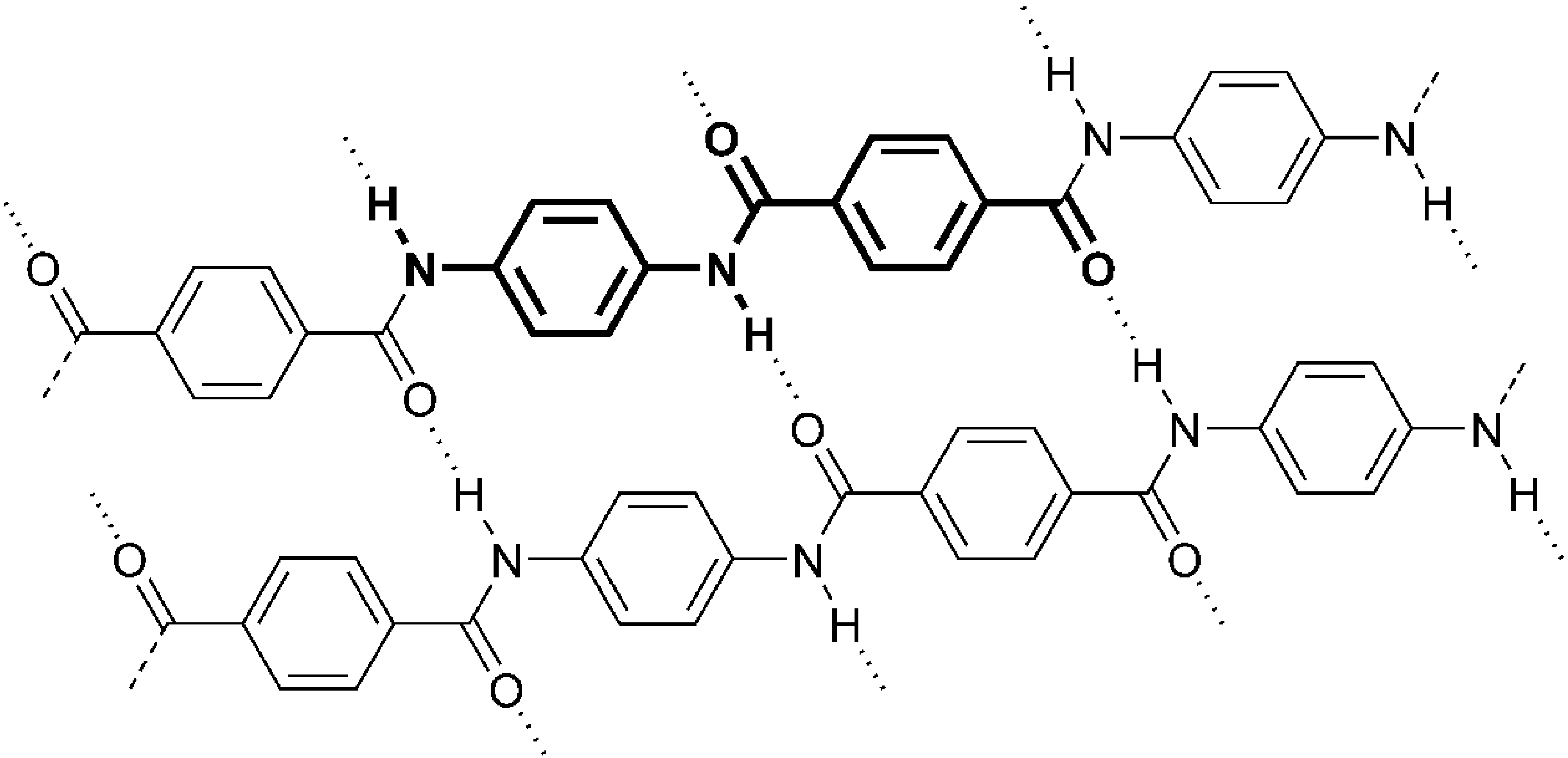


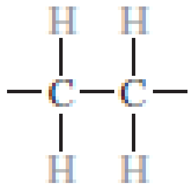
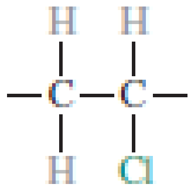
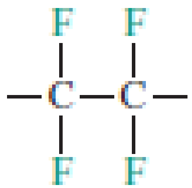
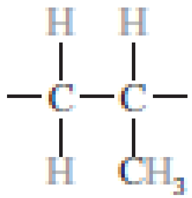
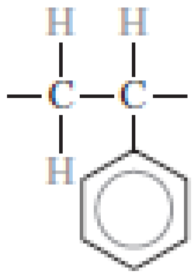
Figure 15.14 The influence of degree of crystallinity and molecular weight on the physical characteristics of polyethylene. (From R. B. Richards, "Polyethylene—Structure, Crystallinity and Properties," *J. Appl. Chem.*, **1**, 370, 1951.)

Kevlar



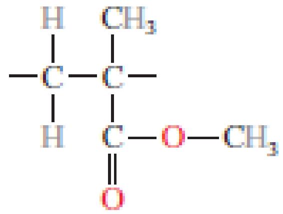
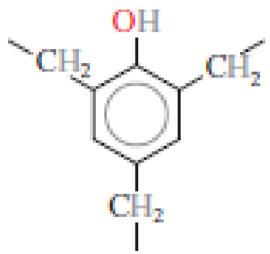
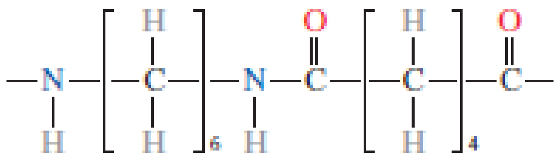
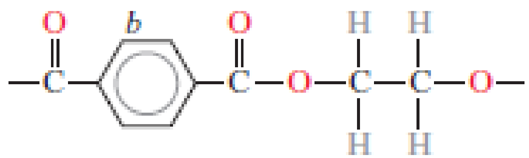
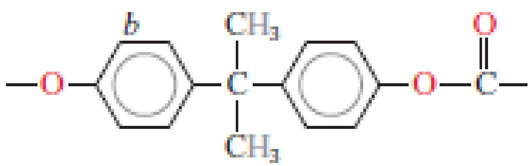
		Kevlar 29	Kevlar 39	Kevlar 149	Acciaio FeB 44K
Density	Kg/m^3	1440	1450	1470	7850
E	GPa	70	140	160	210
TS	MPa	3600	3600	3200	540
Deformation at fracture	%	3.6	1.9	1.5	20
Specific strength	MPa/Kg	2.50	2.48	2.18	0.07

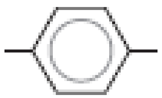
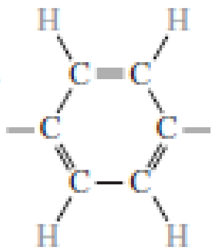
Table 14.3 A Listing of Repeat Units for 10 of the More Common Polymeric Materials

Polymer	*Repeat Unit*
Polyethylene (PE)	
Poly(vinyl chloride) (PVC)	
Polytetrafluoroethylene (PTFE)	
Polypropylene (PP)	
Polystyrene (PS)	

(Continued)

Table 14.3 (Continued)

Polymer	*Repeat Unit*
Poly(methyl methacrylate) (PMMA)	
Phenol-formaldehyde (Bakelite)	
Poly(hexamethylene adipamide) (nylon 6,6)	
Poly(ethylene terephthalate) (PET, a polyester)	
Polycarbonate (PC)	

[b] The  symbol in the backbone chain denotes an aromatic ring as 

Types of polymers

THERMOPLASTICS

Polymers that can undergo shaping upon heating, return to their original state and be reformed indefinitely
They are recyclable.

Ex: Acrylics, Nylons, Polyethylenes

THERMOSETS

When long chain molecules in a polymer become one giant molecule with strong covalent bonds (cross-links) → the polymer is permanently set.

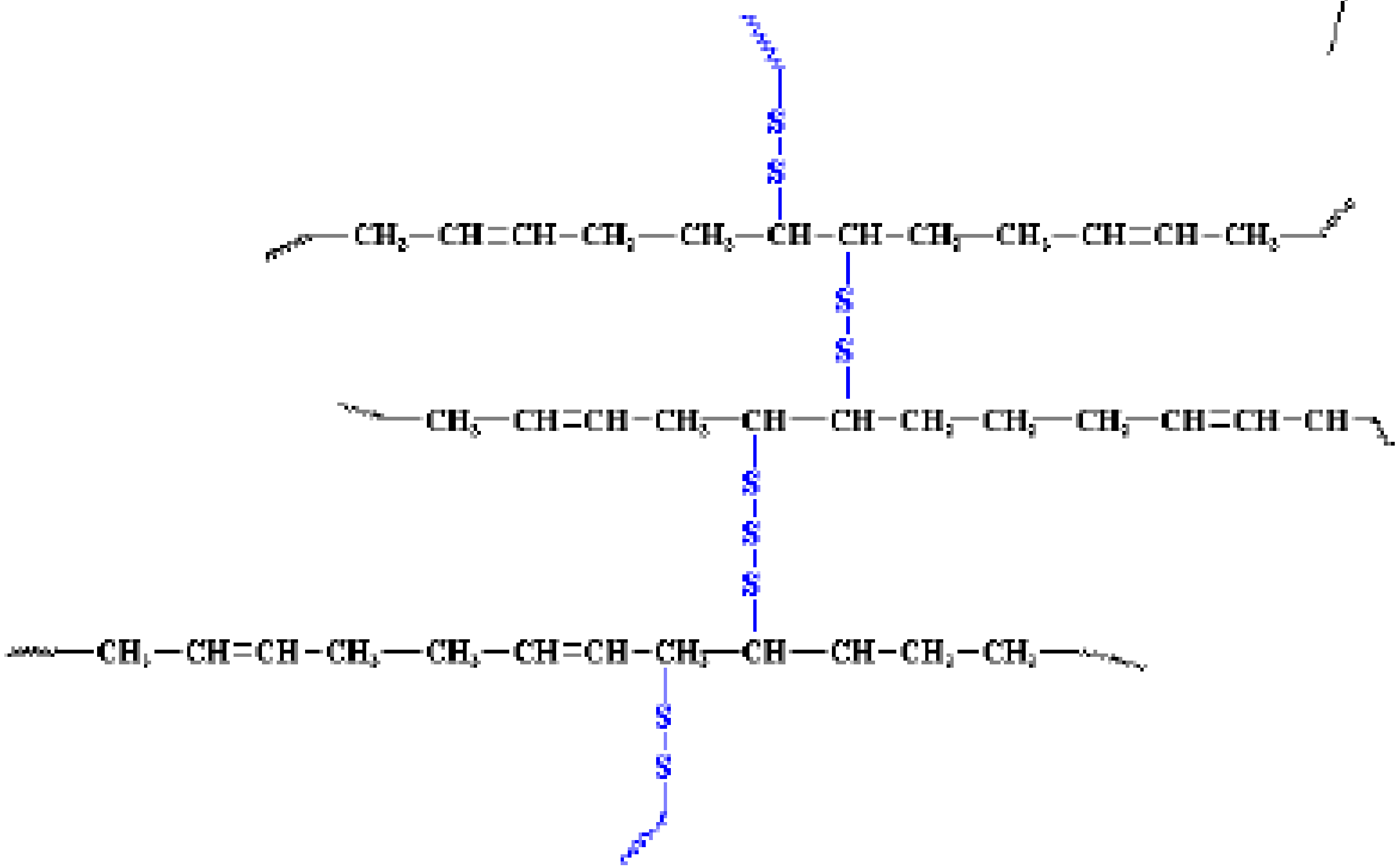
These polymers can be formed only the first time; e.g.: resins, adhesives

The curing reaction of a thermoset is irreversible (unlike thermoplastics)

No T_g value

ELASTOMERS

Linear polymers in which a low number of bridging bonds are introduced, which give the material a three-dimensional structure and ensure very high elastic properties (up to 800% elastic deformation). These bonds are introduced after the molding of the material, through a chemical reaction



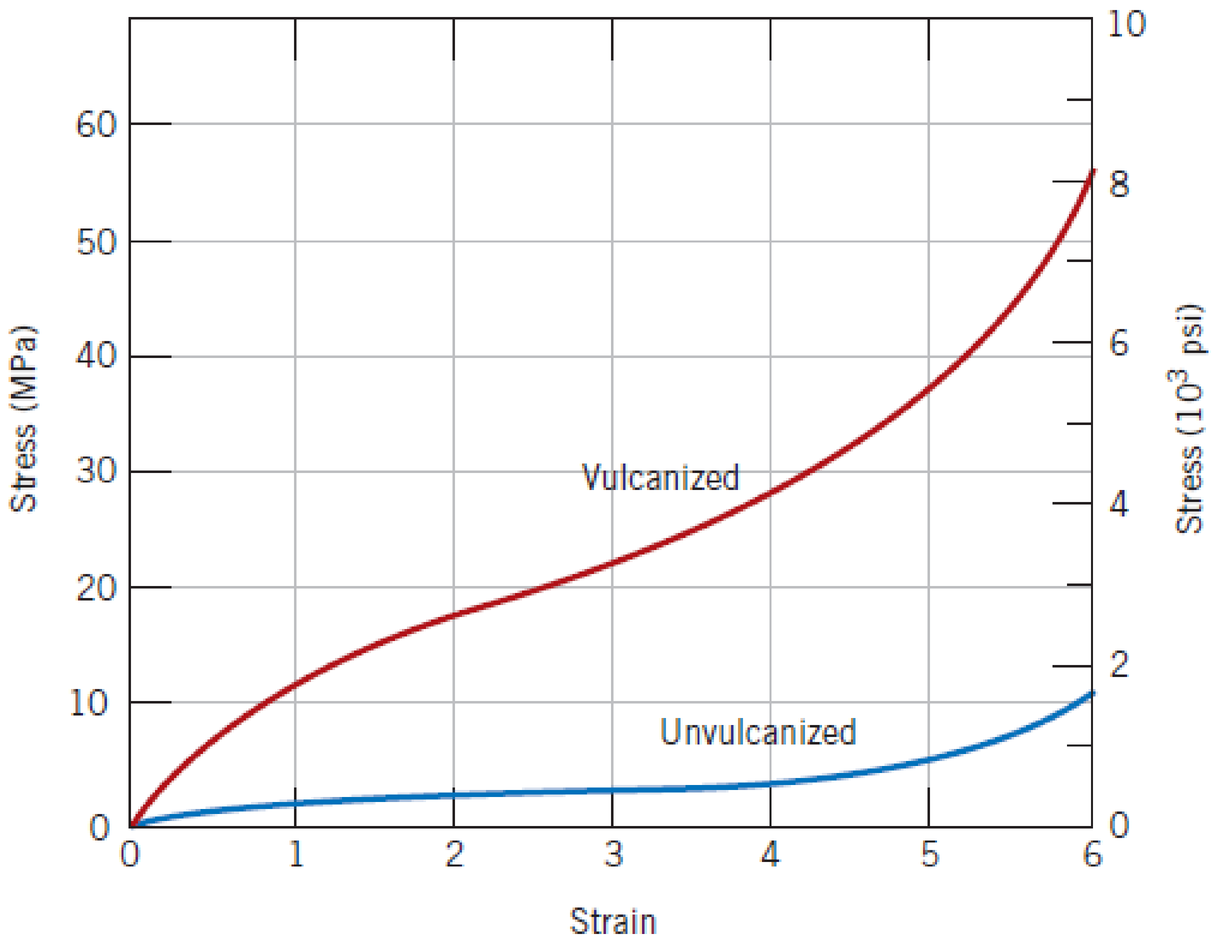
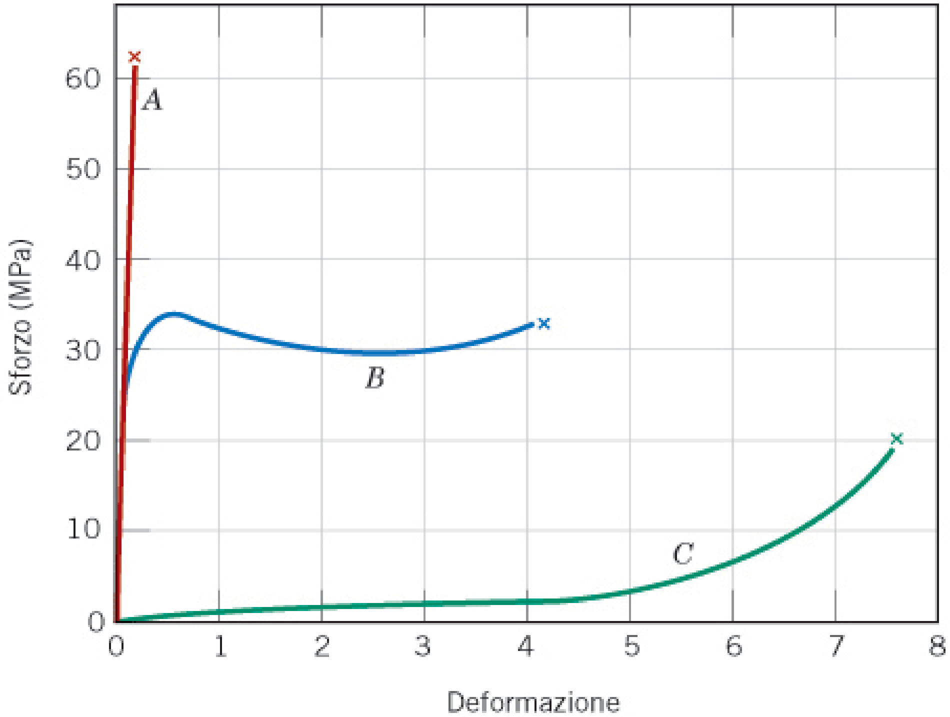


Figure 15.16
Stress–strain curves to 600% elongation for unvulcanized and vulcanized natural rubber.

Effect of glass transition temperature (T_g)

Material	*Glass Transition Temperature* [°C (°F)]	*Melting Temperature* [°C (°F)]
Polyethylene (low density)	–110 (–165)	115 (240)
Polytetrafluoroethylene	–97 (–140)	327 (620)
Polyethylene (high density)	–90 (–130)	137 (279)
Polypropylene	–18 (0)	175 (347)
Nylon 6,6	57 (135)	265 (510)
Poly(ethylene terephthalate) (PET)	69 (155)	265 (510)
Poly(vinyl chloride)	87 (190)	212 (415)
Polystyrene	100 (212)	240 (465)
Polycarbonate	150 (300)	265 (510)



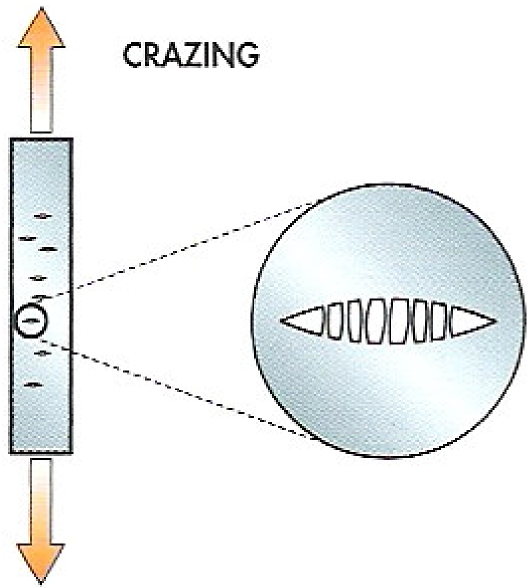
A = termosets or termoplastics below T_g

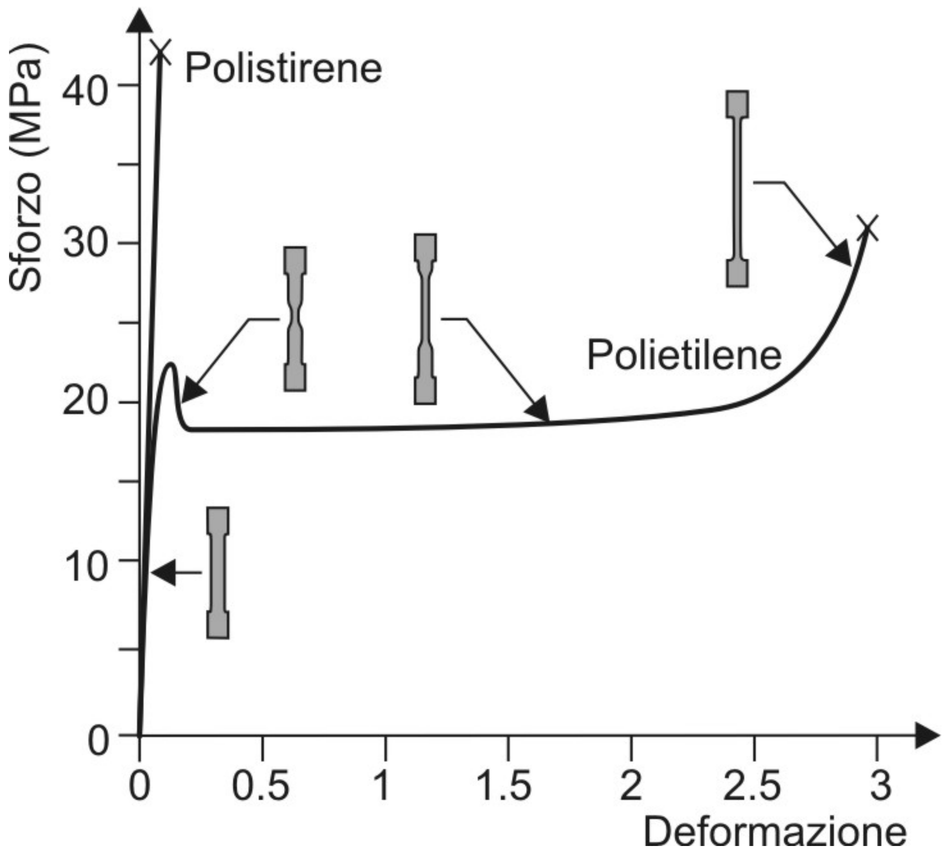
B = termoplastics above T_g

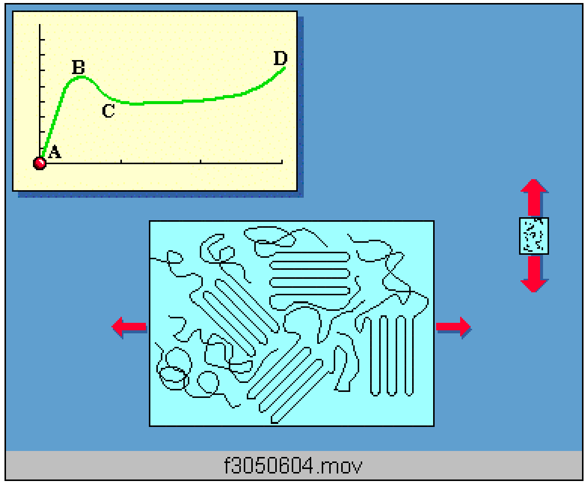
C = elastomer

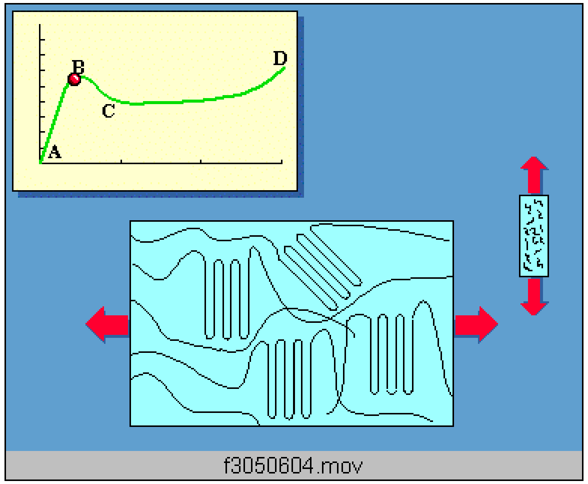
The tensile stress-strain curve for semicrystalline polymers at $T > T_g$:
Upper and lower yield points of the curve are followed by a near horizontal region.
Upper yield point: a small neck forms within the gauge section.
Within this neck the chains become oriented which leads to localized strengthening. There is a resistance to continued deformation at this point and specimen elongation proceeds by the propagation of this neck region along the gauge length.

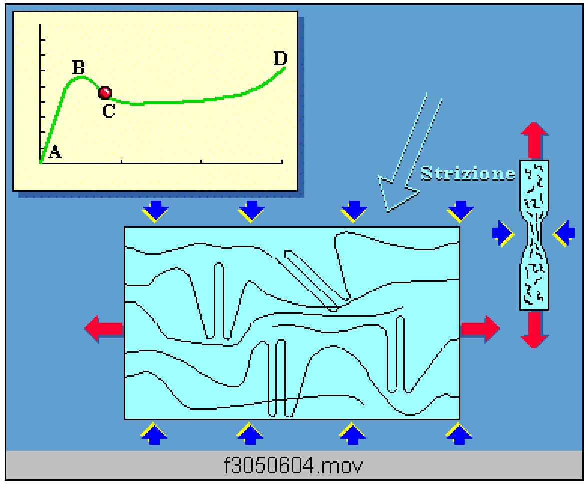
At $T < T_g$ (rigid polymer)

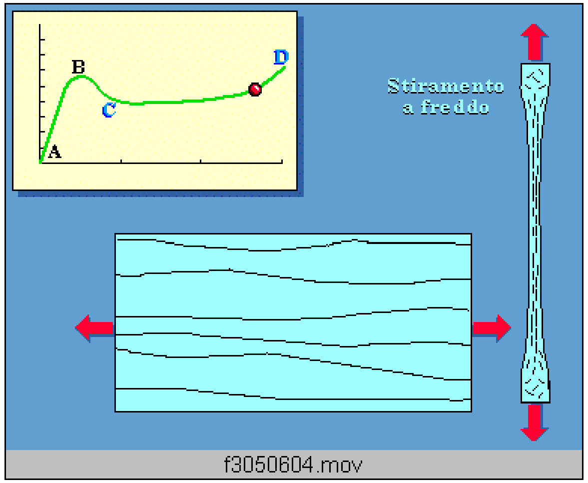


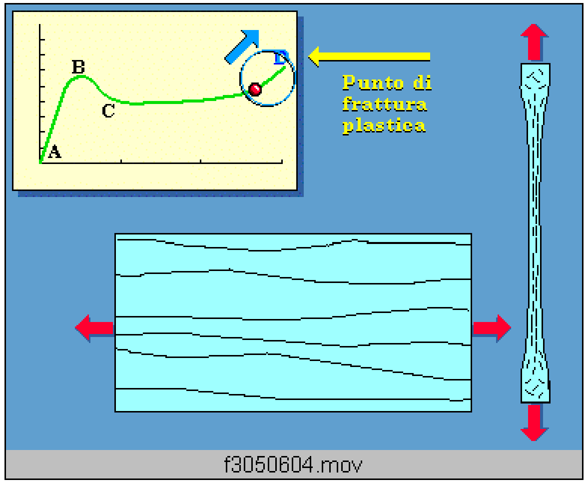




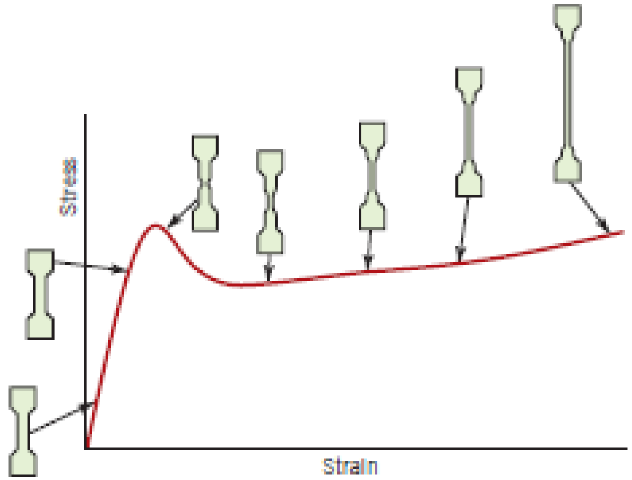








Thermoplastic polymers



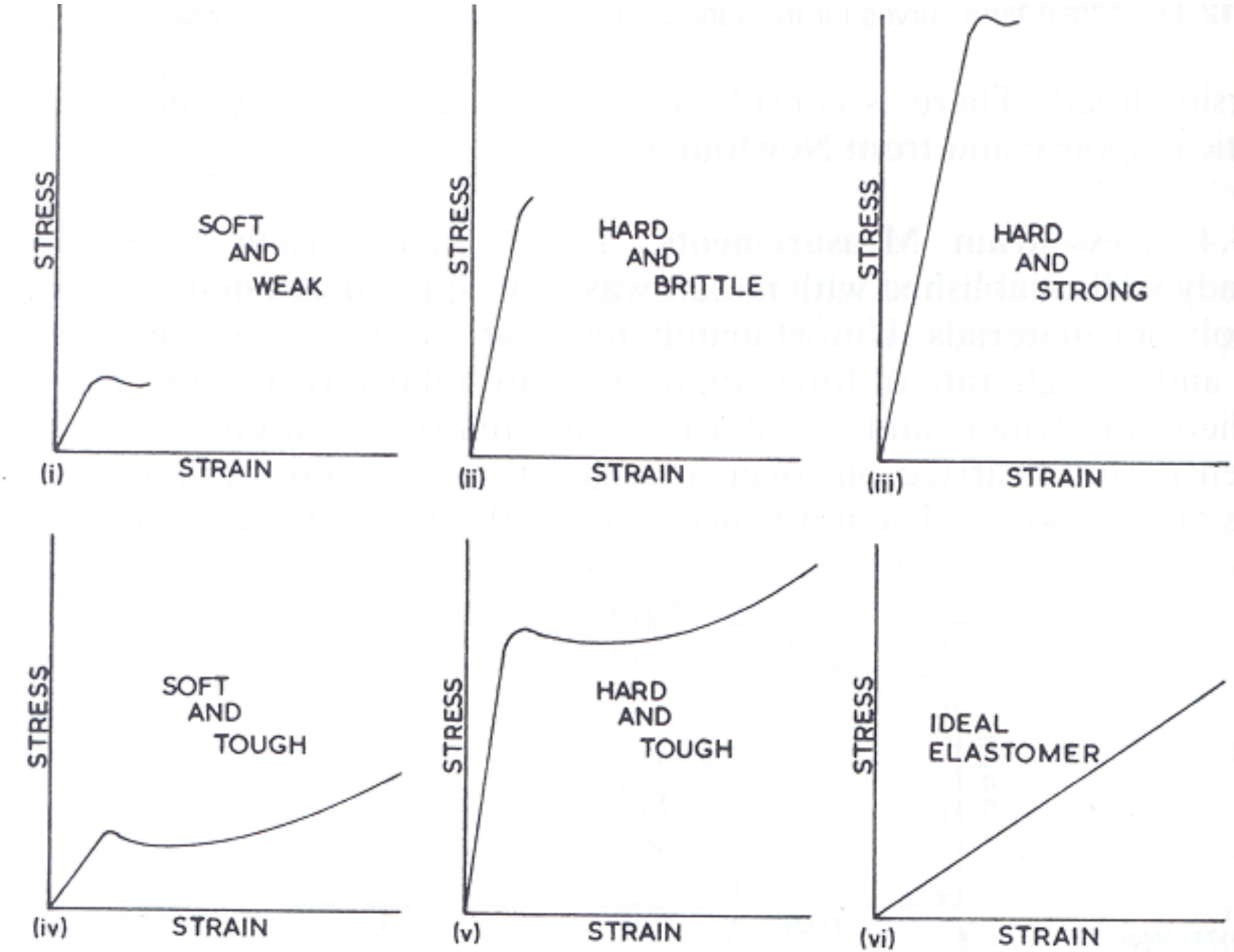
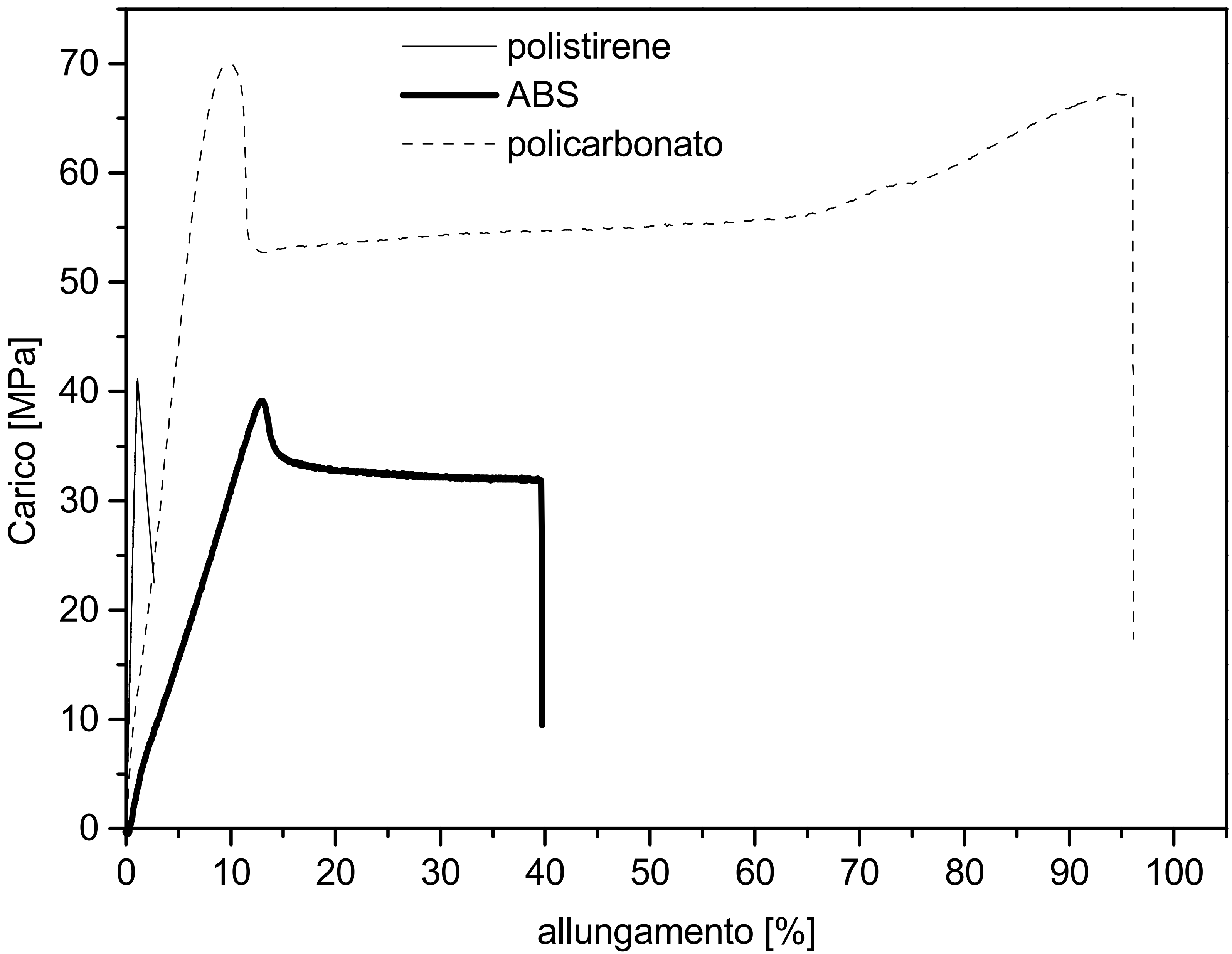


Fig. 12.19 Type of stress-strain diagram for different polymer groups (after Carswell and Nason).


polistirene
ABS
policarbonato
Carico [MPa]
allungamento [%]
0
10
20
30
40
50
60
70
80
90
100

Effect of temperature

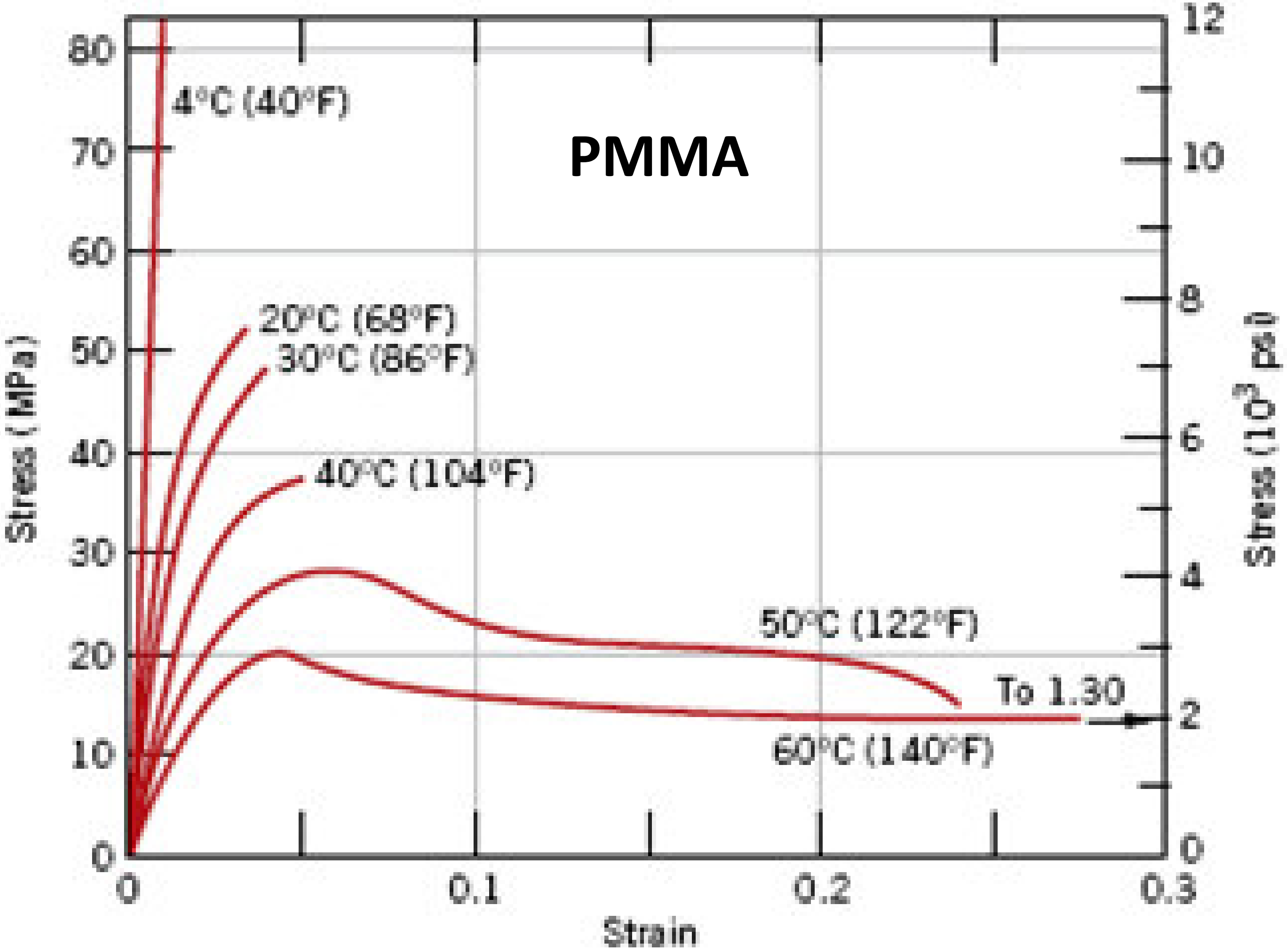


Table 15.1 Room-Temperature Mechanical Characteristics of Some of the More Common Polymers

Material	*Specific Gravity*	*Tensile Modulus [GPa (ksi)]*	*Tensile Strength [MPa (ksi)]*	*Yield Strength [MPa (ksi)]*	*Elongation at Break (%)*
Polyethylene (low density)	0.917–0.932	0.17–0.28 (25–41)	8.3–31.4 (1.2–4.55)	9.0–14.5 (1.3–2.1)	100–650
Polyethylene (high density)	0.952–0.965	1.06–1.09 (155–158)	22.1–31.0 (3.2–4.5)	26.2–33.1 (3.8–4.8)	10–1200
Poly(vinyl chloride)	1.30–1.58	2.4–4.1 (350–600)	40.7–51.7 (5.9–7.5)	40.7–44.8 (5.9–6.5)	40–80
Polytetrafluoroethylene	2.14–2.20	0.40–0.55 (58–80)	20.7–34.5 (3.0–5.0)	—	200–400
Polypropylene	0.90–0.91	1.14–1.55 (165–225)	31–41.4 (4.5–6.0)	31.0–37.2 (4.5–5.4)	100–600
Polystyrene	1.04–1.05	2.28–3.28 (330–475)	35.9–51.7 (5.2–7.5)	—	1.2–2.5
Poly(methyl methacrylate)	1.17–1.20	2.24–3.24 (325–470)	48.3–72.4 (7.0–10.5)	53.8–73.1 (7.8–10.6)	2.0–5.5
Phenol-formaldehyde	1.24–1.32	2.76–4.83 (400–700)	34.5–62.1 (5.0–9.0)	—	1.5–2.0
Nylon 6,6	1.13–1.15	1.58–3.80 (230–550)	75.9–94.5 (11.0–13.7)	44.8–82.8 (6.5–12)	15–300
Polyester (PET)	1.29–1.40	2.8–4.1 (400–600)	48.3–72.4 (7.0–10.5)	59.3 (8.6)	30–300
Polycarbonate	1.20	2.38 (345)	62.8–72.4 (9.1–10.5)	62.1 (9.0)	110–150

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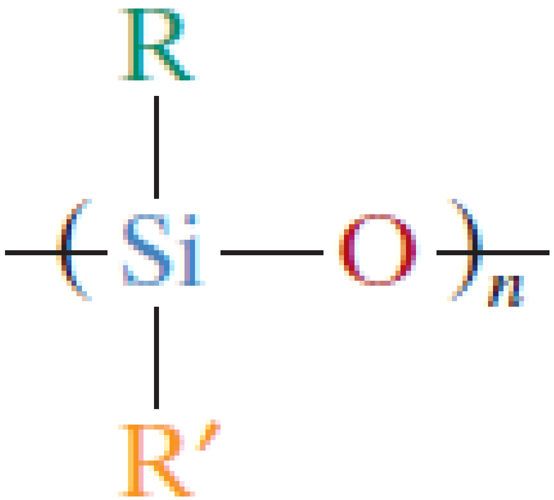
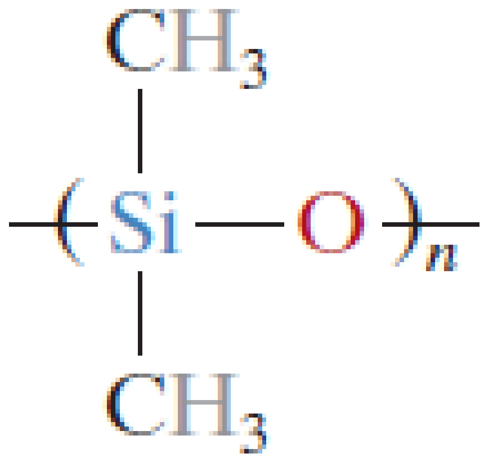
Silicone-based polymers





Processing of polymeric materials

Thermoplastic materials processing technologies:

• Molding, injection

• Thermoforming

• Extrusion

• Calendaring

• Blowing

Processing technologies of **thermosetting** materials:

• Injection molding + transfer (for polymer-matrix composites)

• Compression

• Pultrusion (for polymer-matrix composites)

• Rotational molding

Moulding

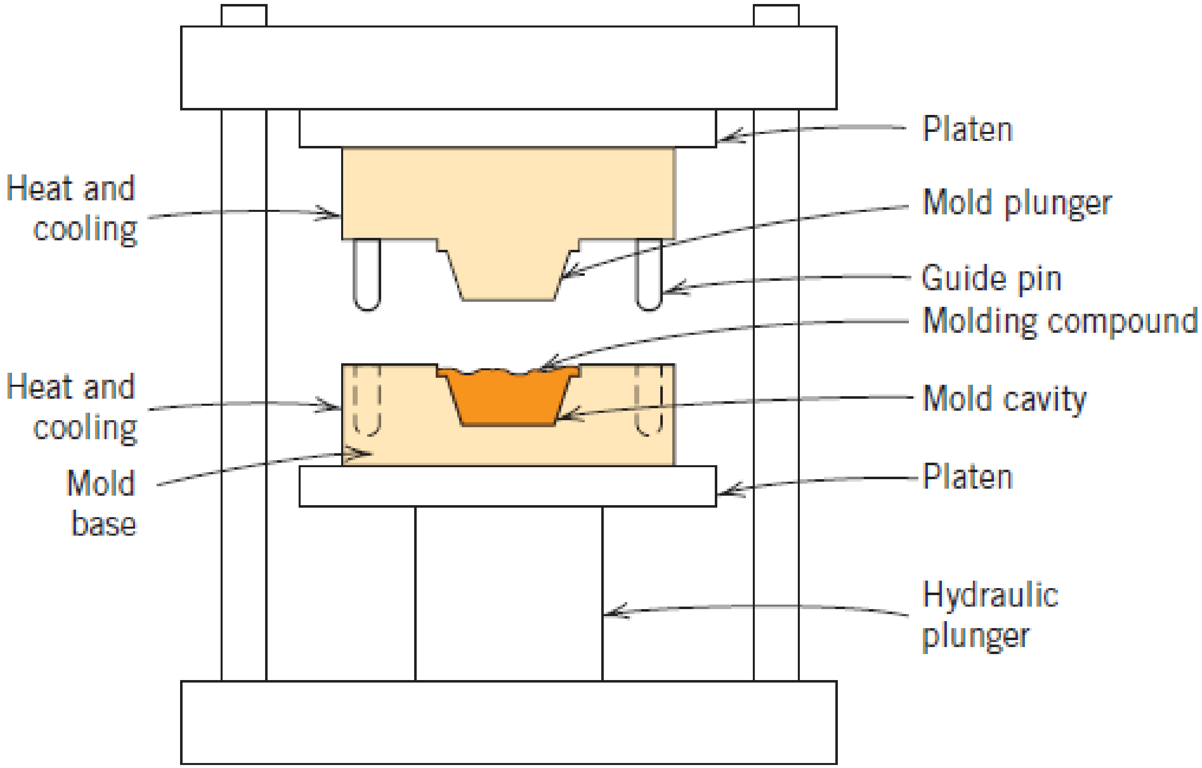
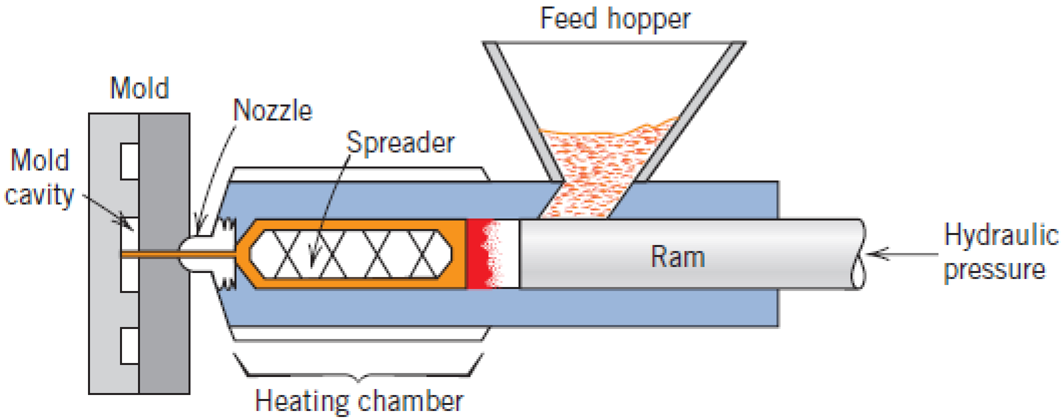


Figure 15.23 Schematic diagram of a compression molding apparatus. (From F. W. Billmeyer, Jr., *Textbook of Polymer Science*, 3rd edition. Copyright © 1984 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

Injection moulding

Figure 15.24 Schematic diagram of an injection molding apparatus. (Adapted from F. W. Billmeyer, Jr., *Textbook of Polymer Science,* 2nd edition. Copyright © 1971 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)



Extrusion

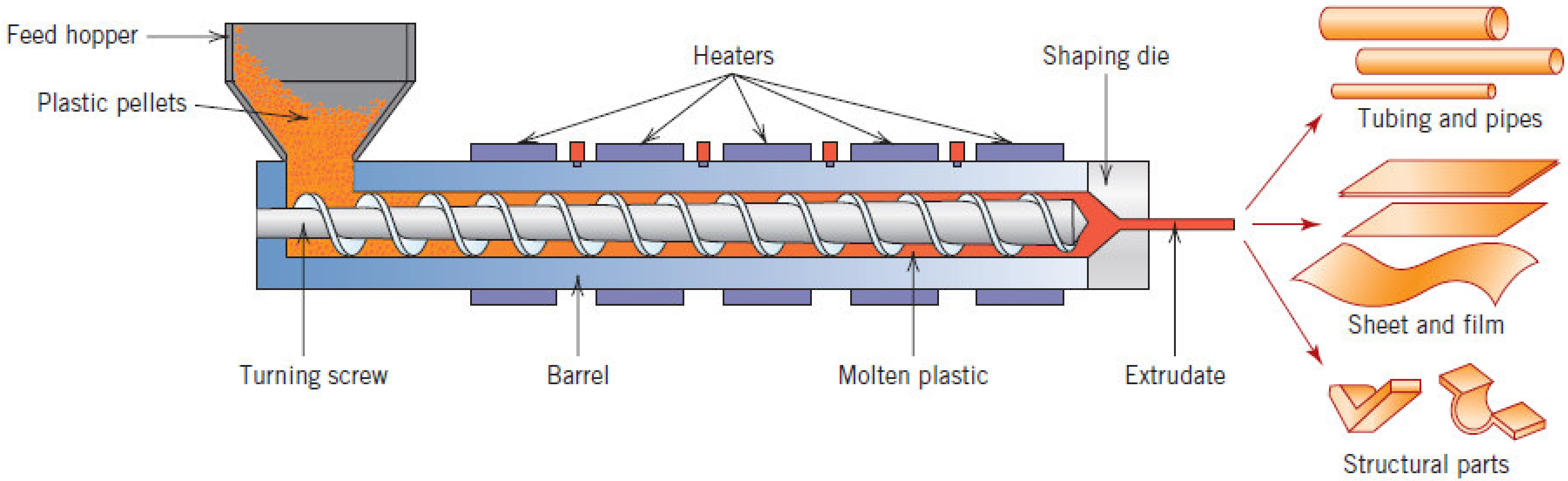


Figure 15.25 Schematic diagram of an extruder. (Reprinted with permission from *Encyclopædia Britannica*, © 1997 by Encyclopædia Britannica, Inc.)

Calendaring (polymer lamination)

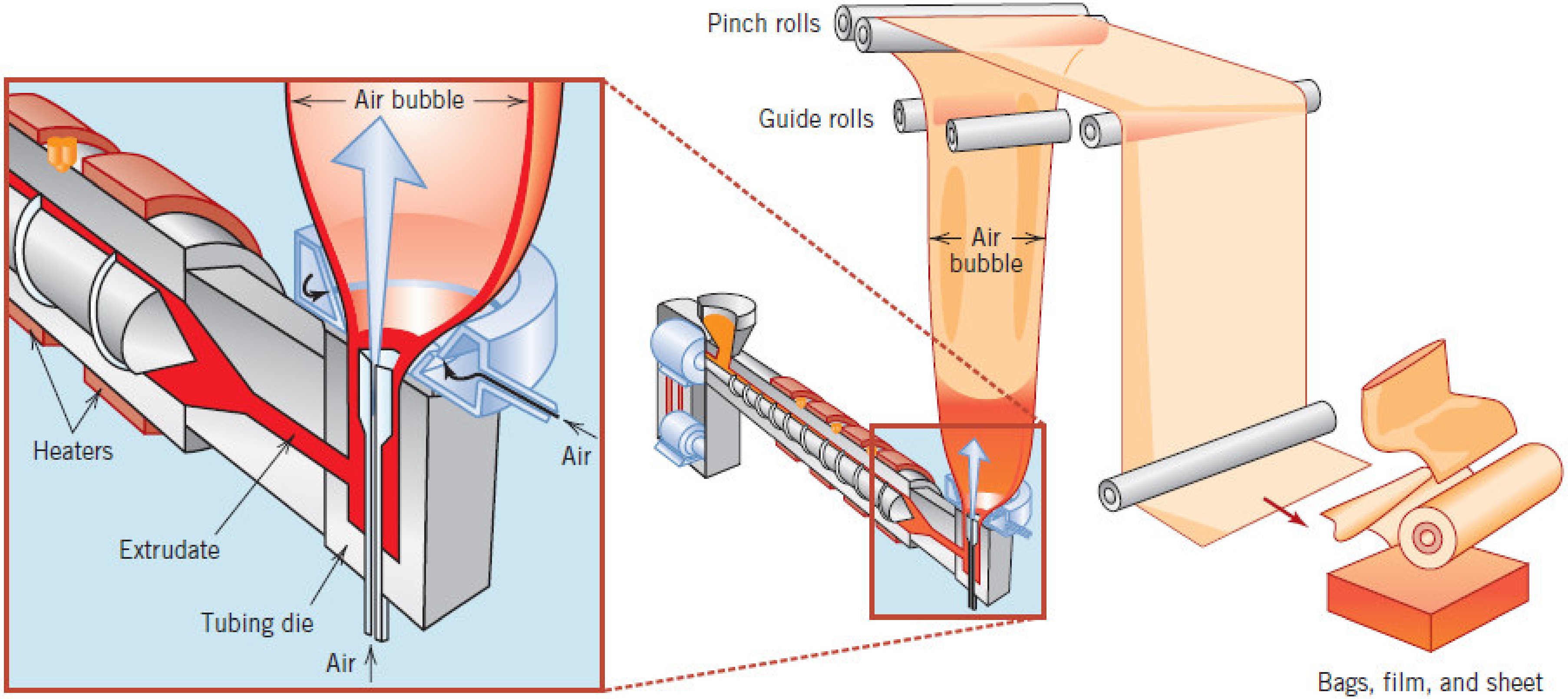


Figure 15.26 Schematic diagram of an apparatus that is used to form thin polymer films. (Reprinted with permission from *Encyclopædia Britannica*, © 1997 by Encyclopædia Britannica, Inc.)

Blowing

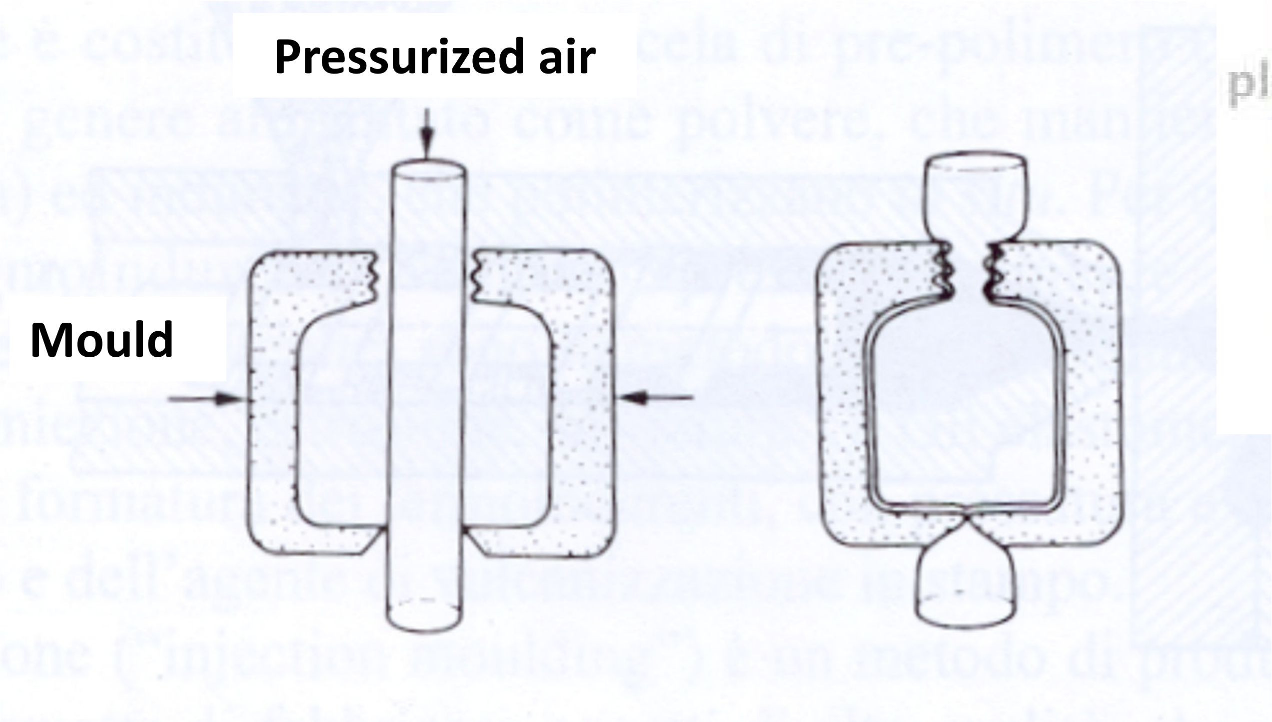
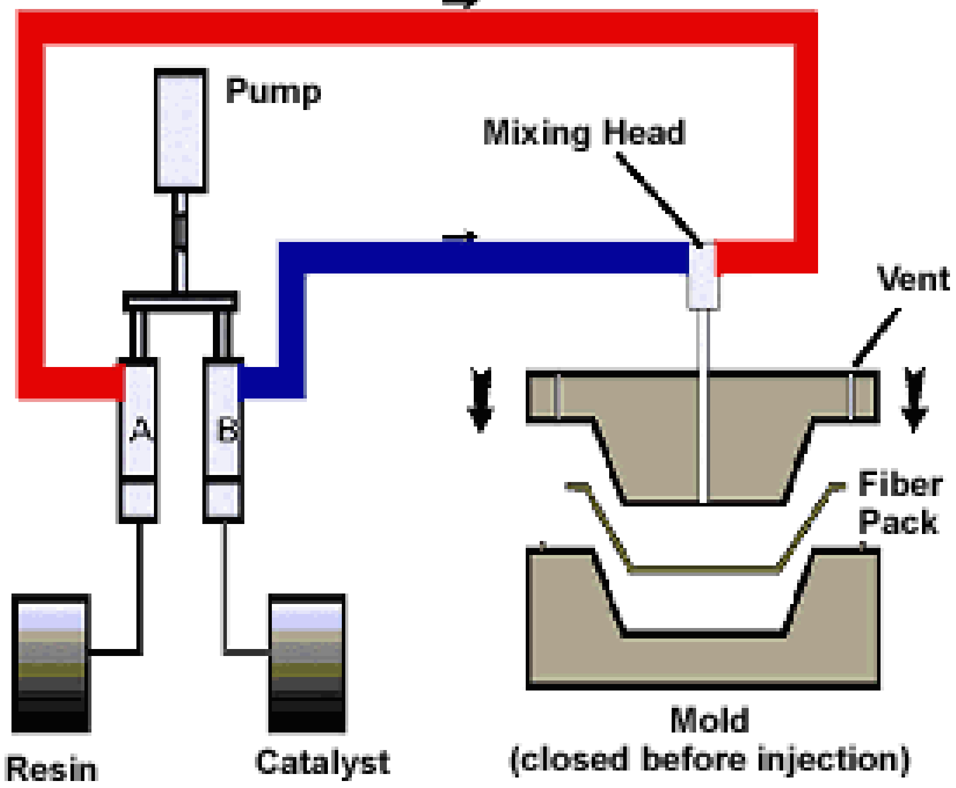


Figura 5.17 – Soffiatura di un termoplastico

RESIN TRANSFER MOLDING (for polymer-matrix composites based on thermosetting polymers)

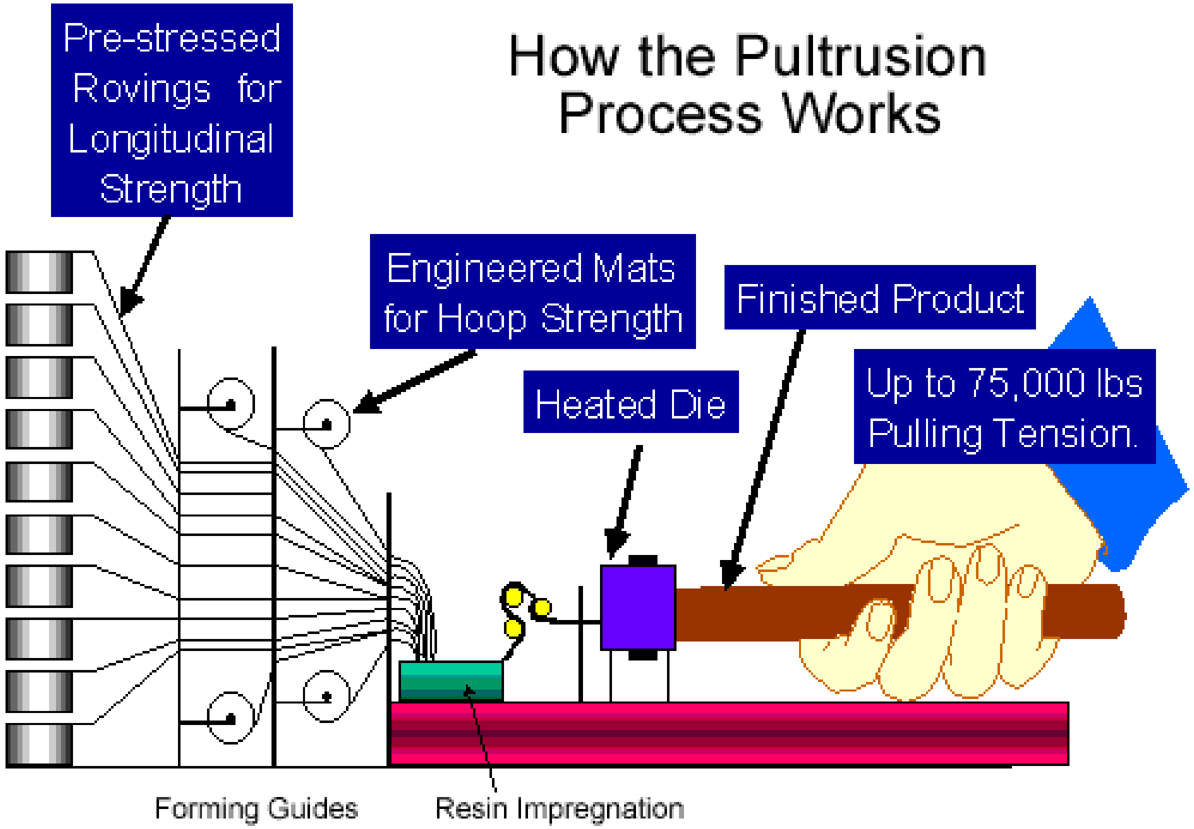
- Resin transfer molding is a manufacturing method that is quite similar to injection molding where plastic is injected into a closed mold

- In the RTM process the preform (precut piece(s) of reinforcement) is placed in the mold, the mold is closed and the thermoset plastic matrix is injected into the mold, once the matrix is cured the part is ejected



PULTRUSION (for polymer-matrix composites based on thermosetting polymers)

- Similar to extrusion of metallic parts
- Pultrusion involves pulling resin-impregnated glass strands through a die
- Standard extruded shapes can easily be produced such as pipes, channels, I-beams, etc.



Degradation of polymers

Breakage of chemical bonds due to the effect (also combined) of:

- Temperature
- Photo-oxidation and UV radiation
- Micro-organisms
- Contact with low MW substances (liquid solvents)

Among the materials used in engineering, **polymeric materials are the most sensitive to T**.

Over time, polymers can undergo changes in their structure and properties that can limit/compromise their use.

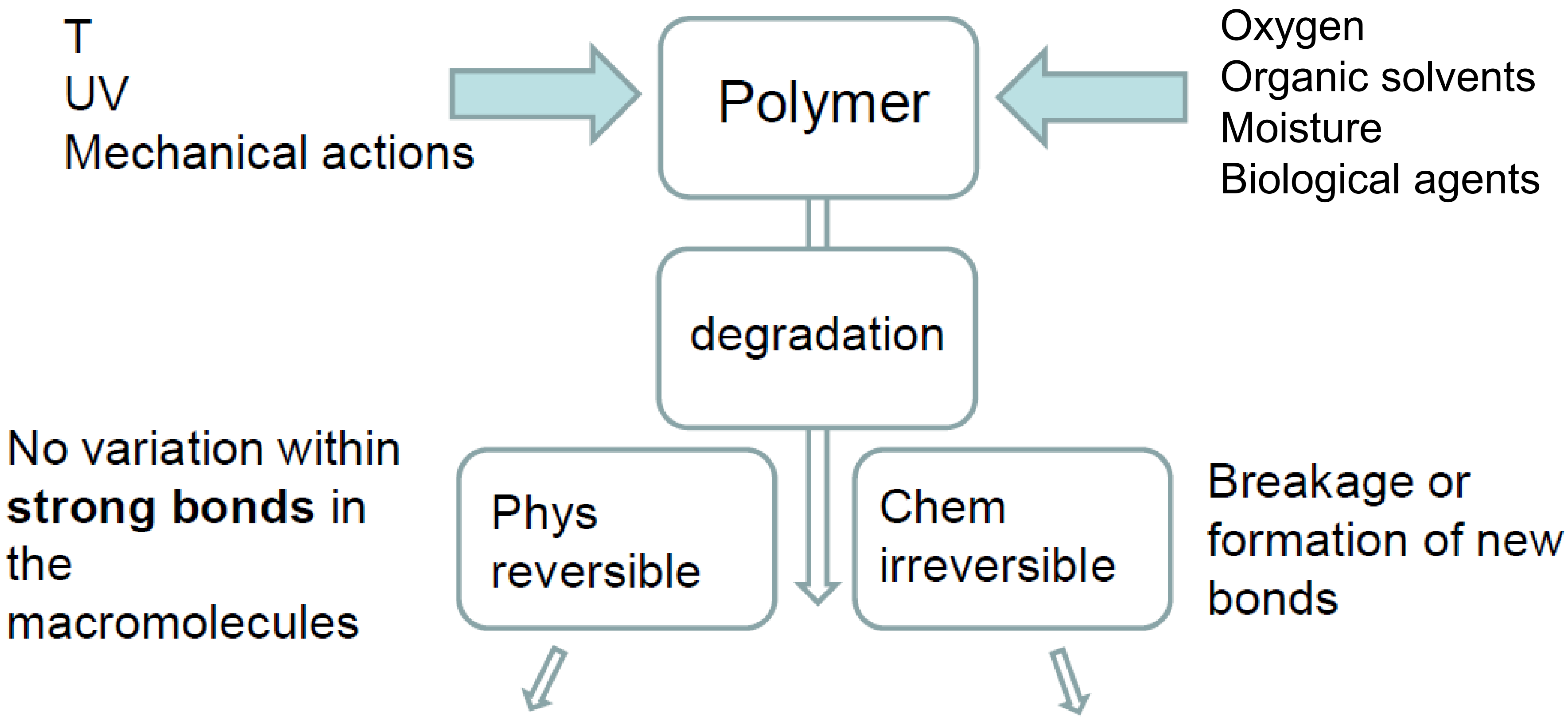
Macroscopic effects of degradation:

• Colour alteration (opaque polymers change their colour and optically-transparent polymers turn yellowish)

• Loss of flexibility

• Loss of toughness

Note that degradation can be caused by:
- environmental agents
- loss of substances with low MW (plasticizers, stabilizers etc.) → exudation



- Thermal transitions (T_g, T_m)
- Aging
- Interactions with liquids (e.g. water)
- Environmental stress crazing (surface micro-cracking)

- Thermal degradation
- Oxidation and UV oxidation
- Interaction with liquids
- Degradation by microorganisms

Fire

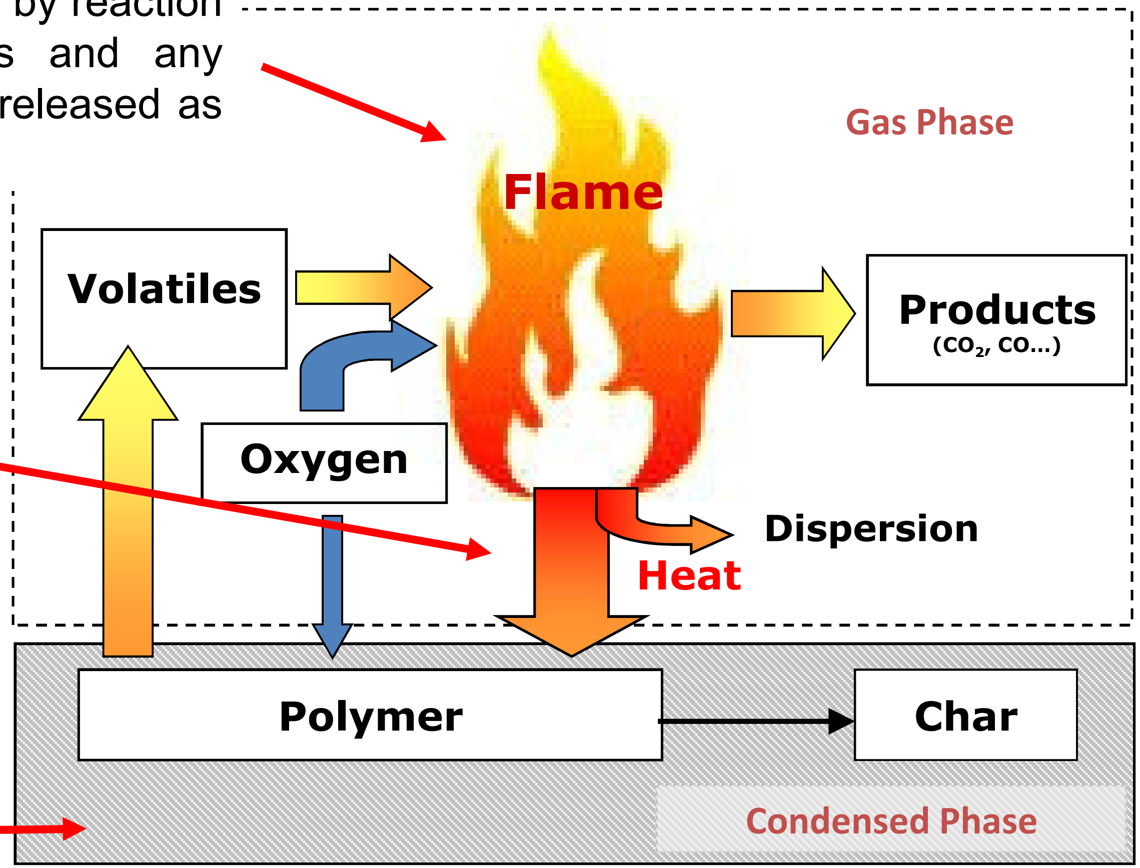
Risks related to the fire:
- reaction to fire
- release of toxic gases (= fumes)

Polymers are flammable materials → **thermal degradation**

Zone of combustion of volatile gases by reaction with O_2, the combustion products and any volatile gases that do not react are released as fumes

Transfer zone of:
- heat downward
- volatile substances upward

Pyrolysis zone: macromolecules are degraded producing volatile substances; some components (e.g. fillers) do not burn and leave a solid deposit of C residues.



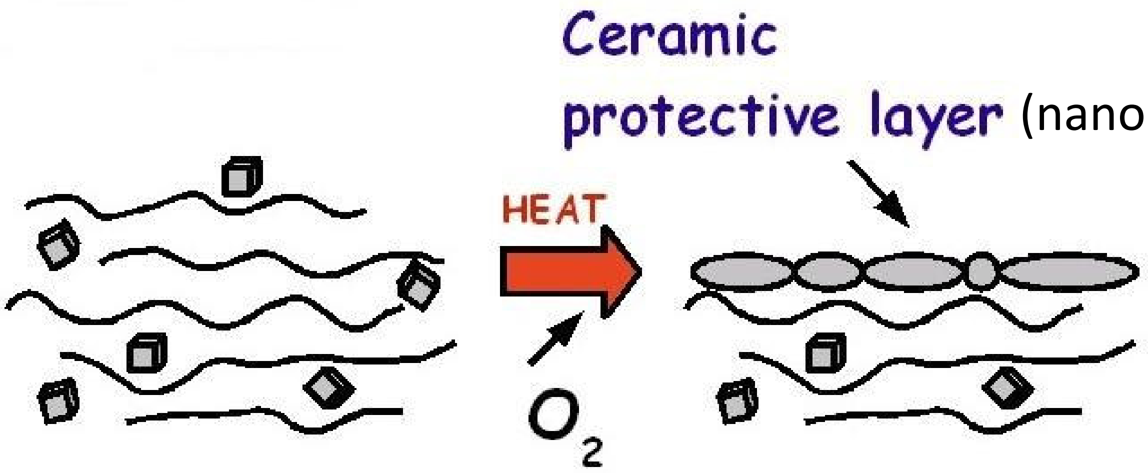
Flame retardants are additives that aim at:

- increasing the resistance of the material to ignition and/or
- reducing the flame propagation speed.

• Inorganic fillers or additives that do not burn and increase the amount of carbon residues on the surface in order to:

- separate the material from the flame,
- reduce the heat that reaches the polymer
- ultimately retard its thermal degradation

• Substances that give endothermic reactions or hydrated compounds in which the evaporation of water takes heat away from the material.



(nanoceramics, CNTs)